CS 1710 Final Project Process Book

Process Book Link:
https://docs.google.com/document/d/1GXfNF4bWgscX5wiARjYBzM7HAS7TAw8cqNDUaye4oz4/edit?usp=sharing

Milestone 1 | Team Formation (9/11/26)

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Milestone 2 | Map (9/12/26)

Topic Brainstorming

- Artificial Intelligence — World Cloud of Opinions on AI over time⭐
- Policy — ICE on Campus, AI in class, and Tech Disparity (Income Inequality, too)
- Cost of Living affects the Poor & the Wealthy differently ⭐
- Varying income across states, dollar worth less in wealthier places
- International individuals arriving to the US, how their age upon arrival affects their future possible wealth
    - Studies have shown moving to a wealthier neighborhood have people more likely to become wealthier themselves ⭐
- Tying National-scale data with Global-scale data of poorer individuals having little access to technology and underdeveloped nations following suit ⭐⭐
- Minorities with Race, Sexual Orientation, Immigrants, etc
- **Comparing wealth/income of CEO's of companies vs their workers** ⭐⭐
- Looking into how Location affects the likelihood of attending college (or other higher education) ⭐

Audience

Possible Audience(s):

- Senior Executives / Workers of Large Billion dollar companies
- Economic Journalists or investors
- Labor Union Representatives
- Democrats
- College graduates looking into what company to work for
- Tech workers or job seekers

Chosen Target Audience: Labor Union Representatives. These individuals can be from a variety of different companies, it doesn't matter, yet they all share the same goal of improving the working conditions and wages for those that work at these corporations. More than likely, they already know that the wealth/income gap between them and their executives/CEO's is already huge, but maybe they don't know how much? Nor do they know the true magnitude of that difference (e.g. how much is $100,000—or the median income—vs $1 billion). Besides wanting to know the exact pay gap, their interests may include using our data to campaign and gain more support, showing off how CEO pay likely grows while the median worker wages stay level (not even growing enough to match the local cost-of-living and inflation). Perhaps they also want to compare against industry averages, competitors, or again, their cost-of-living. Since these are representatives, we assume their visualization literacy is moderate, so we need to make our visualizations easy to understand but still convey plenty of information (using terminology they are likely familiar with). The reason we decided Labor Union Representatives as our target audience was because we shared interests in social good as well as the large wealth inequality in the US, and these representatives serve as a perfect way to link our data and visualizations with their message of supporting workers. We wanted nothing more than for our visualizations to be used for the greater good of society and especially for the working class.

Interesting Questions For Our Audience (About Our Data)

1) How has the CEO-to-median-worker pay ratio evolved across both major and minor corporations over the last decade(s)?
2) How does worker wage growth compare to executive pay over the last decade(s)?
3) How many workdays (or hours) does a CEO need to work in order to earn what their median employee earns in an entire year?
4) Which economic sectors (e.g. Retail/Service, Technology, Manufacturing) exhibit the highest pay disparities? Which exhibit the lowest? For each, how does a randomly (or specifically) chosen company compare against their peer/rival companies?
5) How many workdays does a CEO need to work to earn a median employee's annual pay?
6) If a specifically chosen company redistributed their CEO's bonuses evenly across its non-executive workers, what percentage increase would the median employee receive? What if the CEO earned the median, and the remaining income was redistributed similarly?
7) What proportion of CEO compensation comes from salary versus bonuses, stock awards, and other forms of compensation, and how does that affect the CEO-to-worker pay gap?
8) How does CEO-to-worker pay disparity differ between unioned and non-unionized companies?
9) How does the CEO-to-worker pay gap relate to a company's profitability or financial performance?
10) How much purchasing power does the median worker's salary provide compared with the CEO's compensation when adjusted for local cost of living?
11) How does employee turnover or workforce size relate to CEO-to-worker pay disparity across companies?
12) How much of each person's income is consumed by basic necessities compared to their CEO?

Data

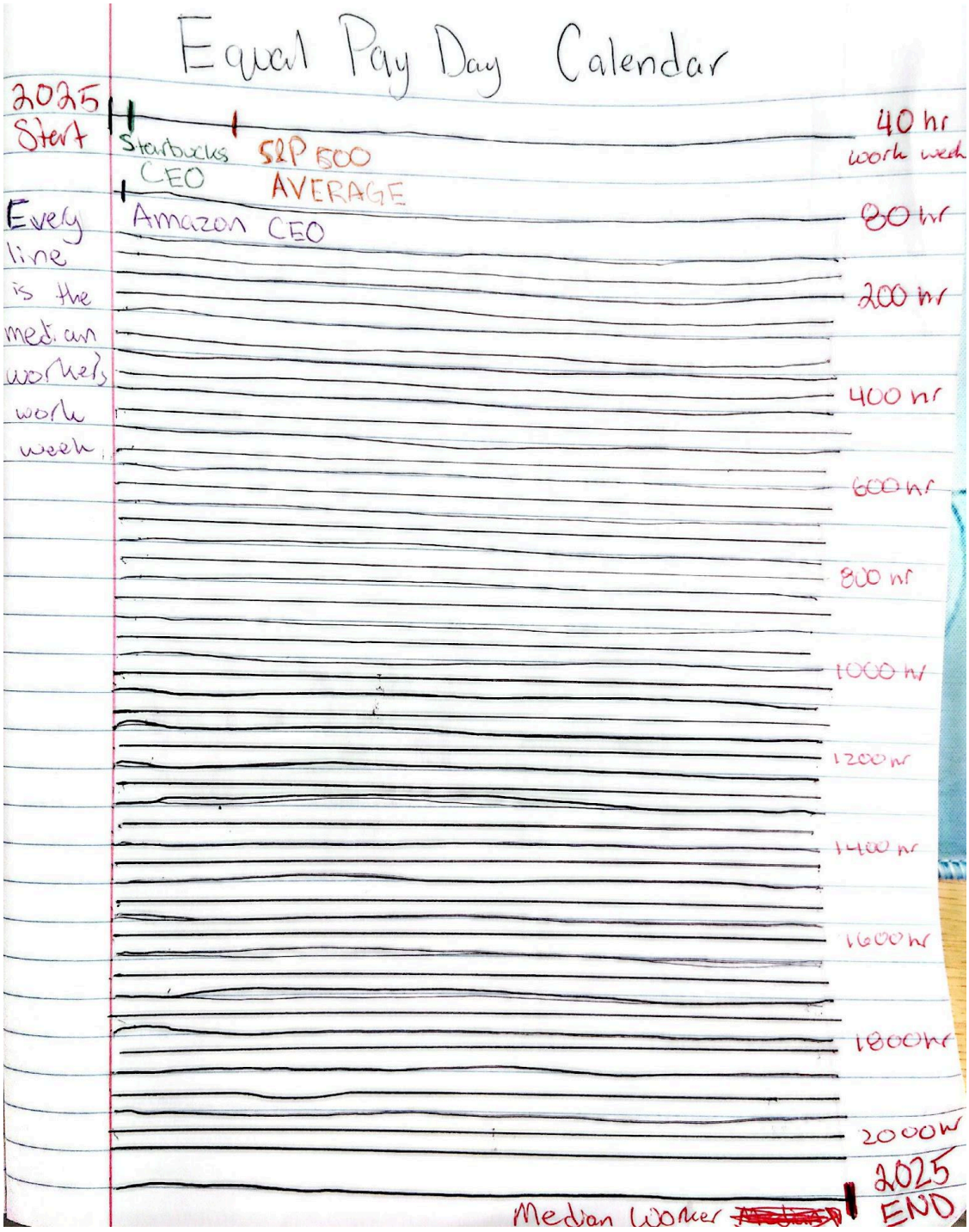
Data We Have/Will Obtain

We do not have any data yet, but there is such a vast amount of data we would need to collect and especially across many different datasets. For example, we need individual corporate financial statements to learn about the pay of executives and the pay of workers, data on the cost-of-living in several different areas, and probably so much more. We would need things like company names (categorical), industry sector (categorical), geographic location, CEO total annual compensation (quantitative) — including base salary, stock awards, bonuses, etc, median employee annual compensation (quantitative), total employee count (quantitative), presence of a union (categorical), regional price parity indices (quantitative), housing costs (quantitative), inflation data (quantitative), and so much more that we have yet to think of (but we see that it is mainly quantitative data).

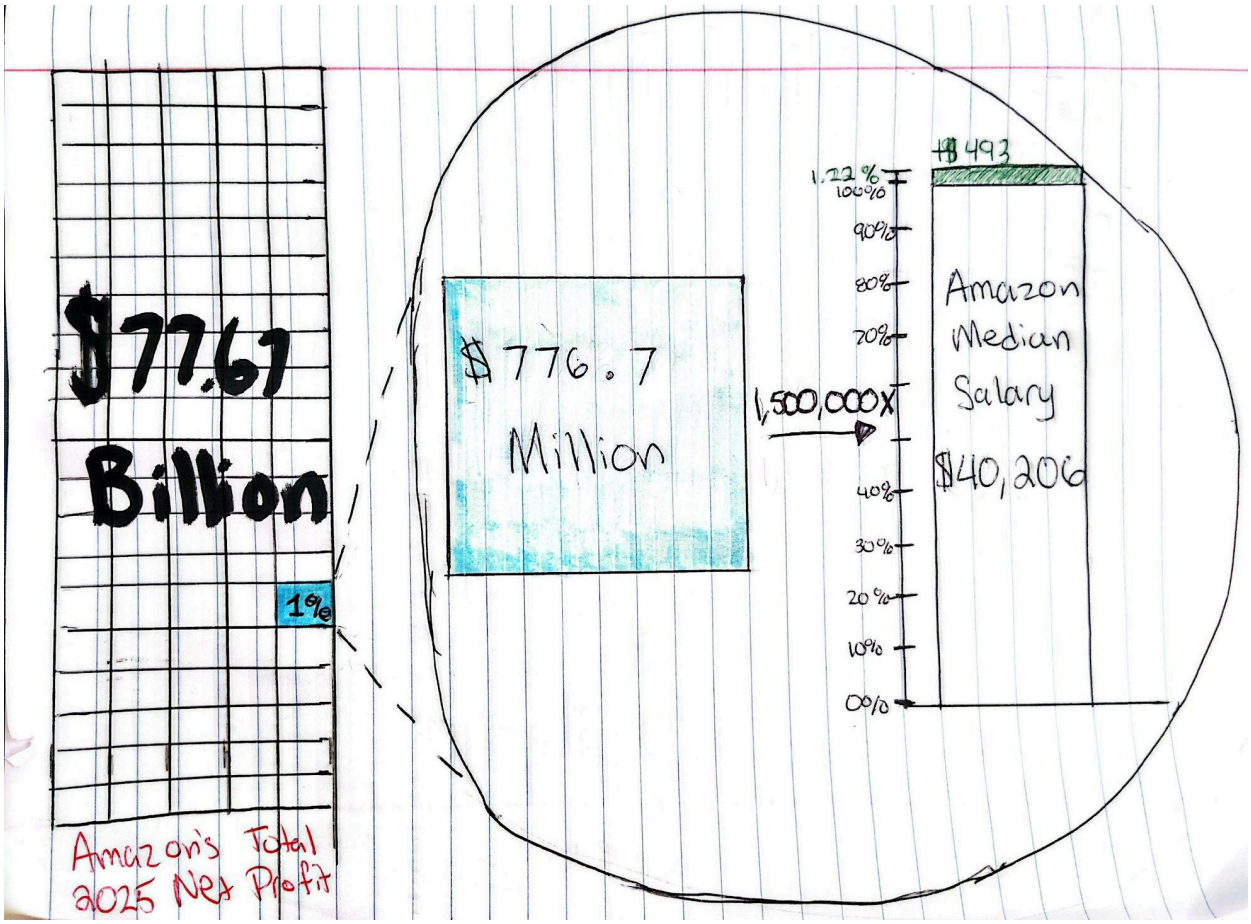
Individual Sketches

Ashle Sanchez

How far into the year does the CEO have to work to match their average/median employee's annual income?



What happens to a median worker's salary if the company they work for redistributes only 1% of its annual profits?



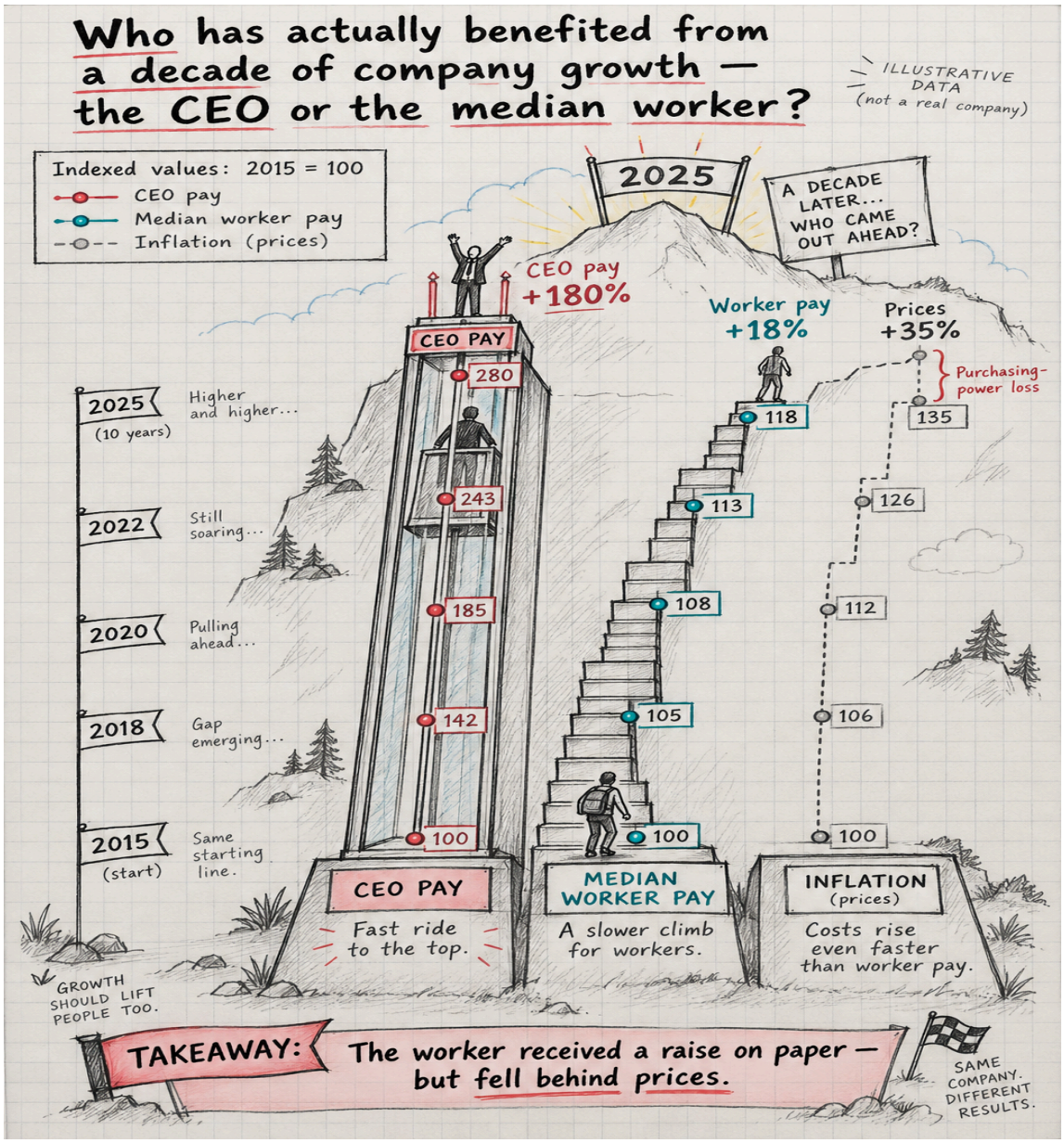
Which economic sectors drive America's largest income disparities?



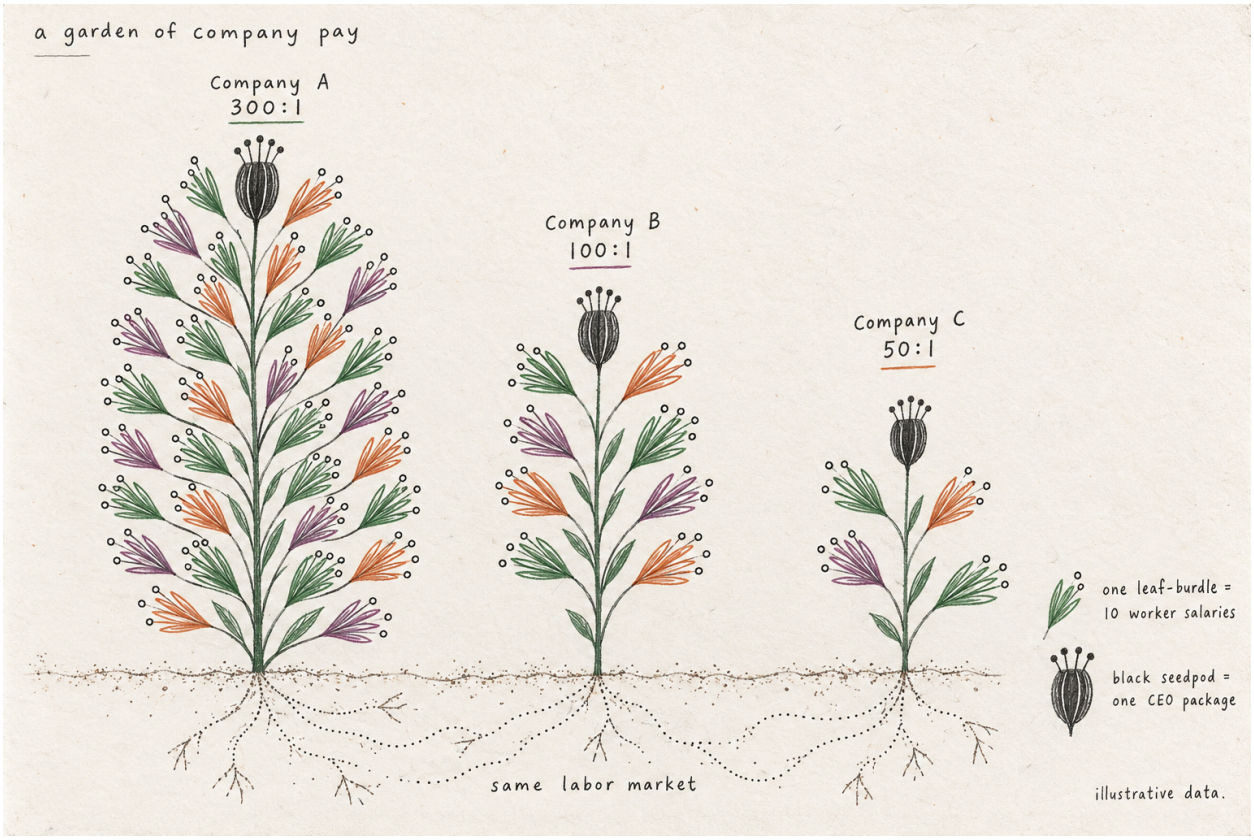
Reflection: Out of my three sketch questions, I took inspiration from the questions our group came up with, more specifically, you can see the inspiration from Question 3 for Sketch 1, Question 5 for Sketch 2, and Question 4 for Sketch 3. However, my sketches were a little more specific based on data that I could find online. My first sketch answered essentially the same question, but was based on a few companies and the average, where certainly more companies could be placed on my sketch. My second sketch changed question 5 significantly because it didn't fit the narrative I wanted to tell, so I switched the point of focus from a CEO's annual income towards the company's profits, and seeing how only 1% could really impact the lives of all the workers. My last sketch shortened question 4, but tried to still answer it based on the information displayed (although more companies could be added onto the sketch). The only reason my sketches needed to slightly alter our questions was mainly because of the data that I could find online about these topics, OR the question was too long and a single sketch was trying to answer too much. I would argue that the new concise questions made after the sketches were better in the sense that they were clear and simple to answer, but our old questions still inspired them in the first place. So, both were great, but maybe just needed to be a little more refined.

Aisha Ahmed

Initial Visualization #1 Name/Question



Initial Visualization #2 Name/Question



Initial Visualization #3 Name/Question

HOW MUCH OF A WORKER'S PAYCHECK SURVIVES THE LOCAL COST OF LIVING COMPARED WITH THE CEO'S?

ILLUSTRATIVE DATA — replace with company and location data.

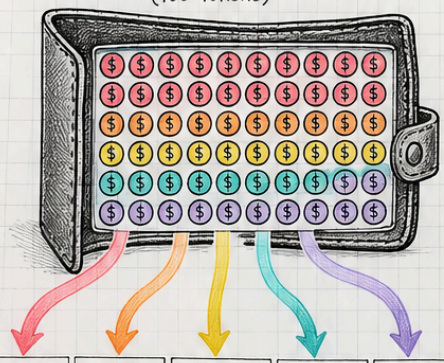
Same city. Same costs. Different reality.

TOKEN COLOR KEY (SAME FOR BOTH)
- = Housing
- = Food
- = Transportation
- = Healthcare
- = Taxes / Other
- = Wealth-building / Discretionary

100 tokens = 1 month of pay (each $ = 1%)

MEDIAN WORKER

Monthly pay: $5,000
(100 tokens)



Housing	Food	Transportation	Healthcare	Taxes / Other
42%	14%	12%	8%	24%

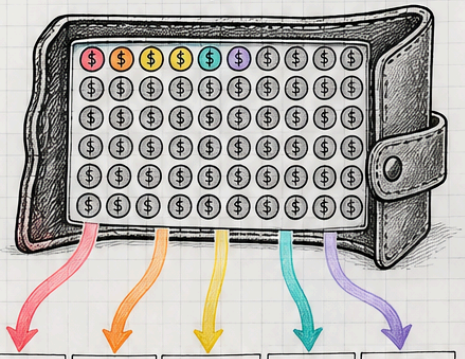
Total: 100%

0% uncommitted
(In this high-cost-city example)

Everyday costs leave little room to breathe.

CEO

Monthly compensation: $500,000
(100 tokens)



Housing	Food	Transportation	Healthcare	Taxes / Other
2%	1%	1%	0%	0%

Necessities total: 4%
(housing + food + transportation + healthcare)

96%

More room for choice.

wealth-building / discretionary

SHARE CONSUMED BY BASIC NEEDS

(housing + food + transportation + healthcare)

Worker: 76%

CEO: 4%

Same city. Same prices. Different pressure.

Same essentials. Very different share of pay.

TAKEAWAY: The same city charges both people — but not the same share of their pay.

Reflection: One paragraph about how the sketch questions compare with the team's original questions? Why do they differ (if they do)? Are some questions better than others? Why? If you stayed with the original questions, why?

My three sketch questions stayed connected to our team's original focus on CEO-to-worker pay inequality, but I made them more specific so that each visualization could communicate one clear idea. The first compares how CEO pay, worker pay, and inflation have changed over time, which helps show that even when workers receive raises, their actual purchasing power may still decline. The second develops this further by comparing how much of a worker's income and a CEO's compensation is consumed by the same basic living expenses. Finally, the third makes the CEO-to-worker pay ratio easier to understand by representing companies as plants, where each bundle of leaves represents multiple worker salaries and the seedpod represents one CEO compensation package. I think these questions work better visually than some of our broader original questions because they translate large or abstract financial figures into familiar experiences and objects. Together, they show not only that a pay gap exists, but also how it grows and what that difference means in workers' everyday lives, which would be especially useful for our audience of labor union representatives.

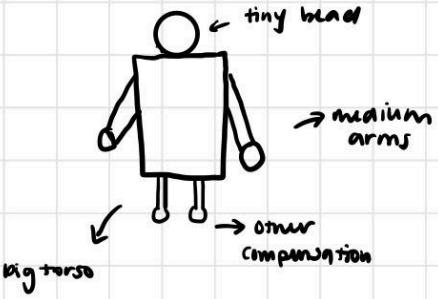
Kayleen Mukire

What proportion of CEO compensation comes from salary versus bonuses, stock awards, and other forms of compensation, and how does that affect the CEO-to-worker pay gap?

CEO compensation as a layered suit / person

- Head = base salary
- Arms = bonuses / incentives
- Torso = stock awards
- Legs = other compensation

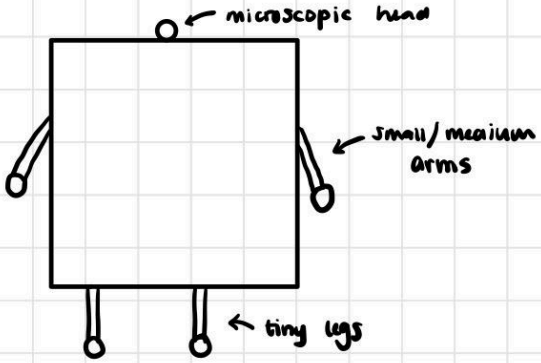
APPLE (Tim Cook)



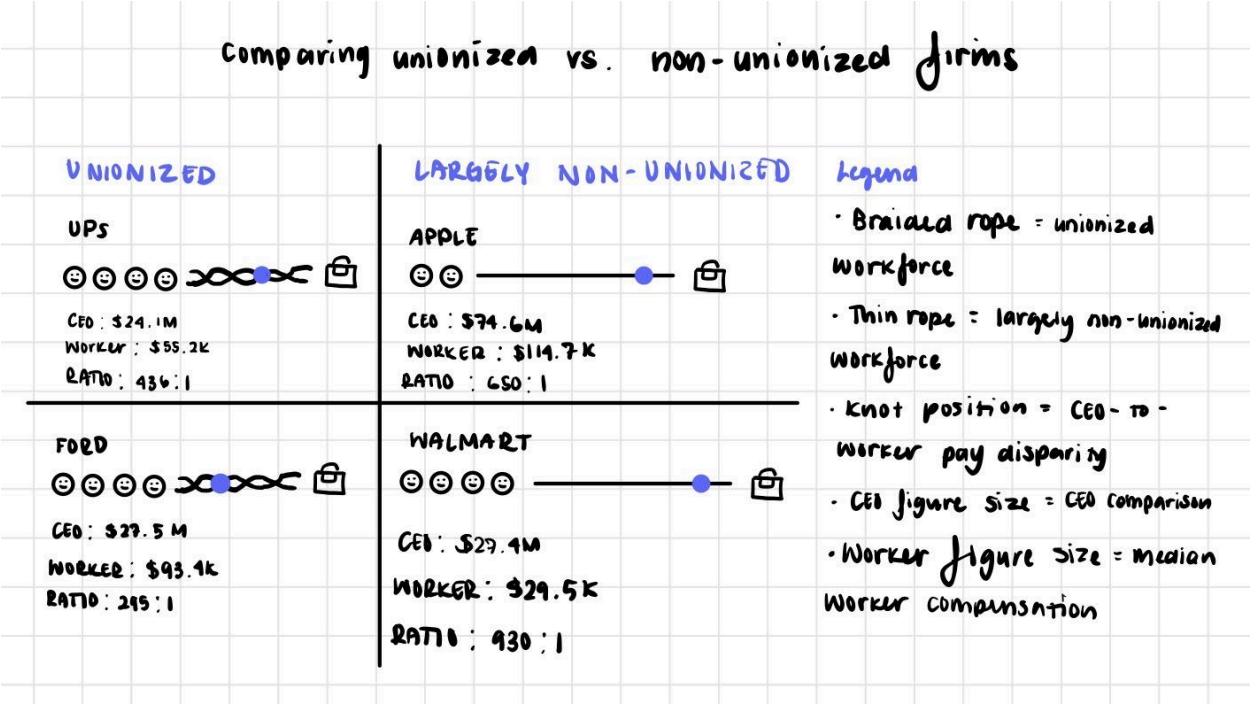
Salary : $3.0 M
Stock Rewards : $58.09 M
Non-equity incentive compensation : $12.0M
Other compensation : $1.52 M
Total : $74.61 M
Median employee compensation : $139,483
2025 CEO-to-worker ratio : 533:1

Salary : $62K
Stock Rewards : $90.3 M
Bonus : $5.0M
Non-equity incentive compensation : $30K
Other compensation : $41BK
Total : ~$95.8 M
Median employee compensation : $14,674
2025 CEO-to-worker ratio : 6,666:1

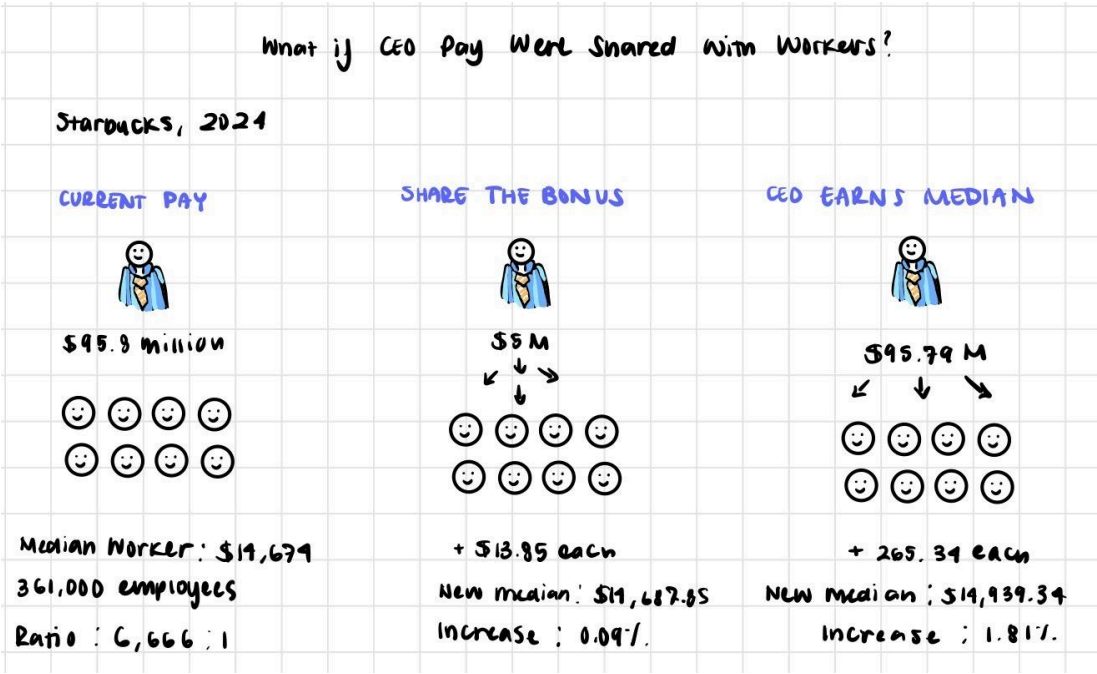
STARBUCKS (Brian Niccol)



How does CEO-to-worker pay disparity differ between unionized and non-unionized companies?



If a specifically chosen company redistributed their CEO's bonuses evenly across its non-executive workers, what percentage increase would the median employee receive? What if the CEO earned the median, and the remaining income was redistributed similarly?



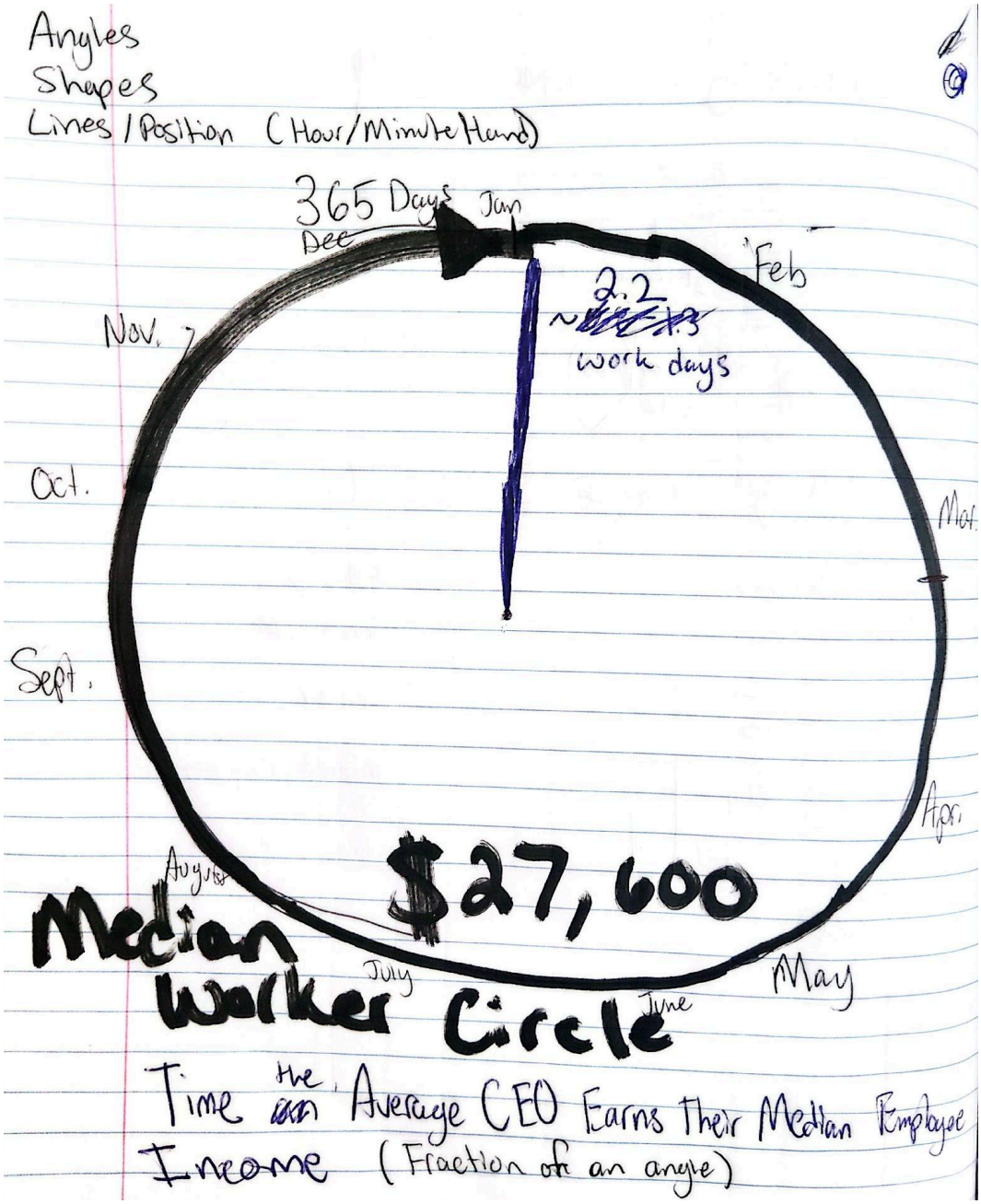
Reflection: Our three sketch questions stayed connected to the team's overall focus on CEO-to-toworker pay inequality, but each approached it differently. One examined how CEO compensation is structured, another compared unionized and largely non-unionized companies, and the third explored what would happen if CEO compensation were redistributed to workers. These questions differ from the team's broader historical questions because they focus more on the structure and consequences of the pay gap. I think they work well because they make the inequality more concrete and relevant to our target audience of labor union representatives, while still supporting the team's original goal of understanding CEO-worker pay disparities.

Milestone 3 | Sketch (9/20/26)

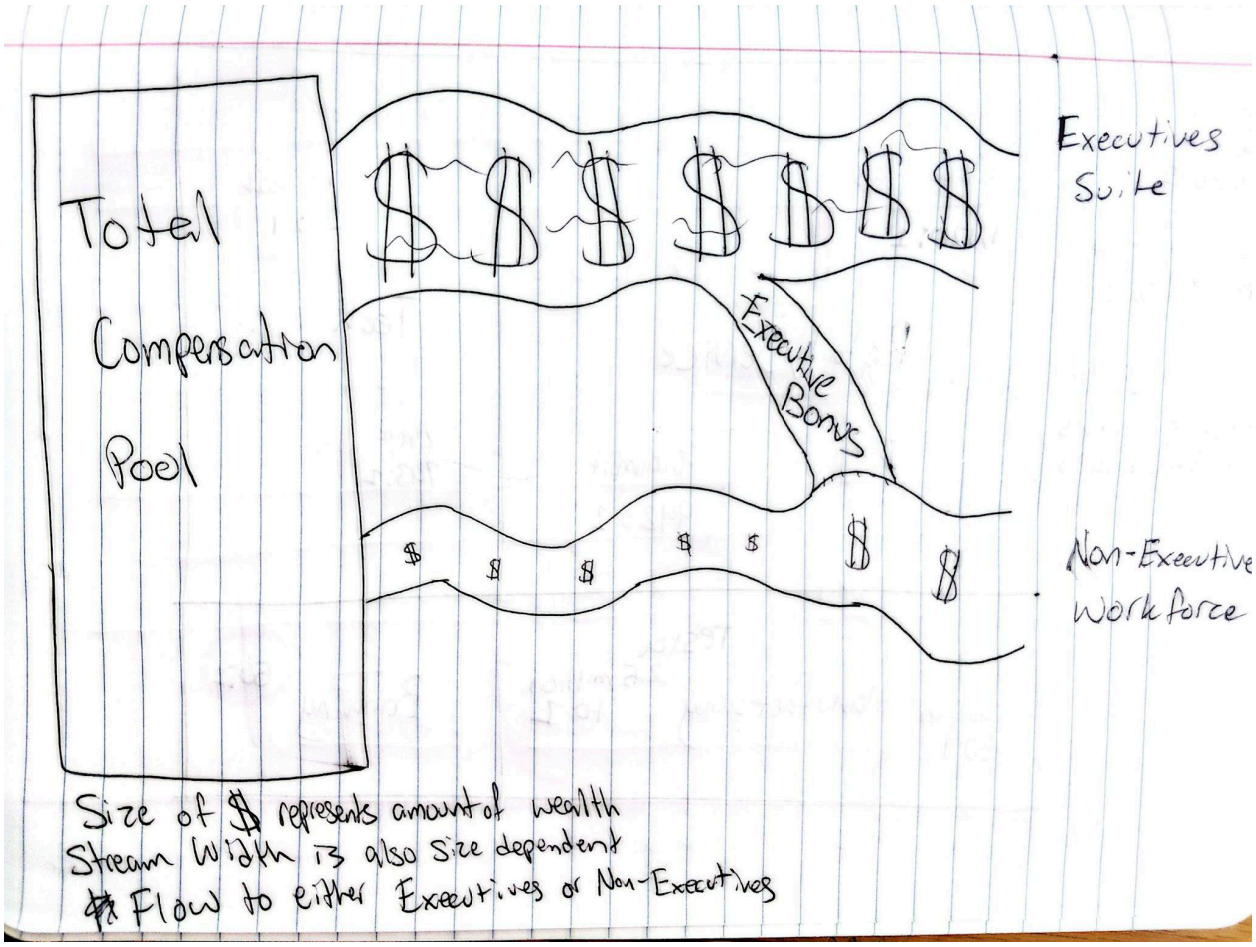
Individual Sketches

Ashle Sanchez

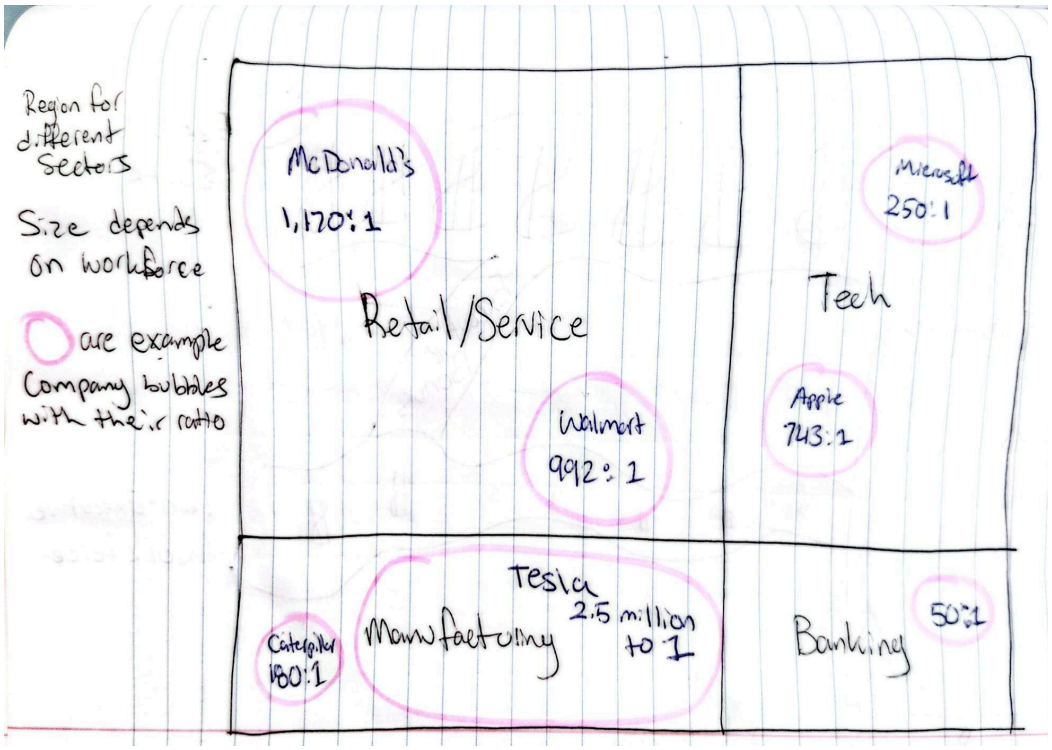
How many workdays (or hours) does a CEO need to work in order to earn what their median employee earns in an entire year?



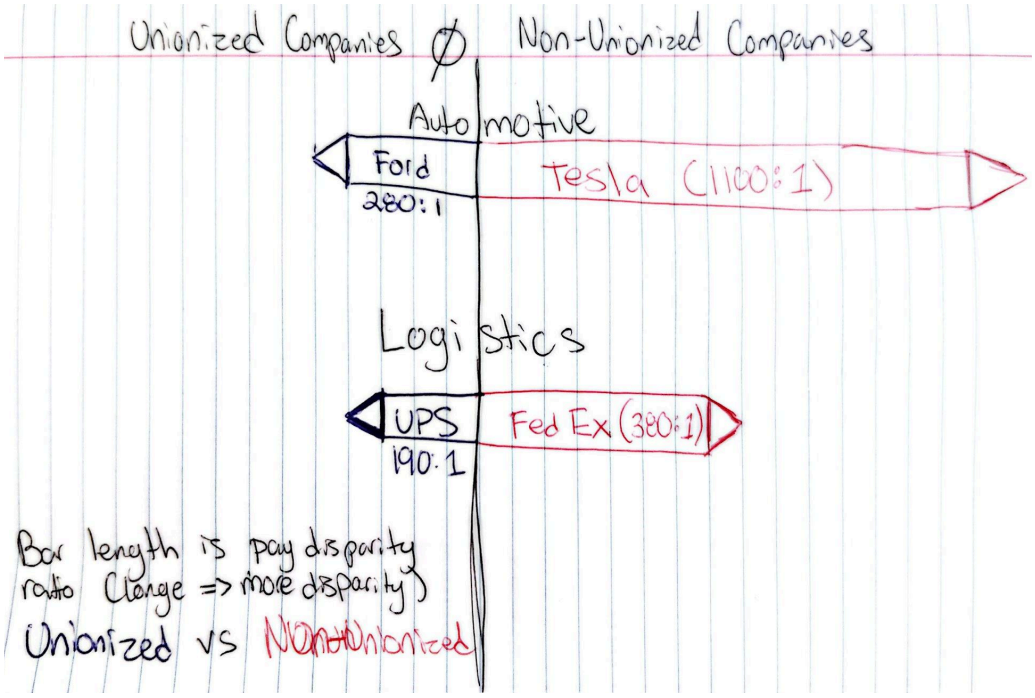
If a specifically chosen company redistributed their CEO's bonuses evenly across its non-executive workers, what percentage increase would the median employee receive?



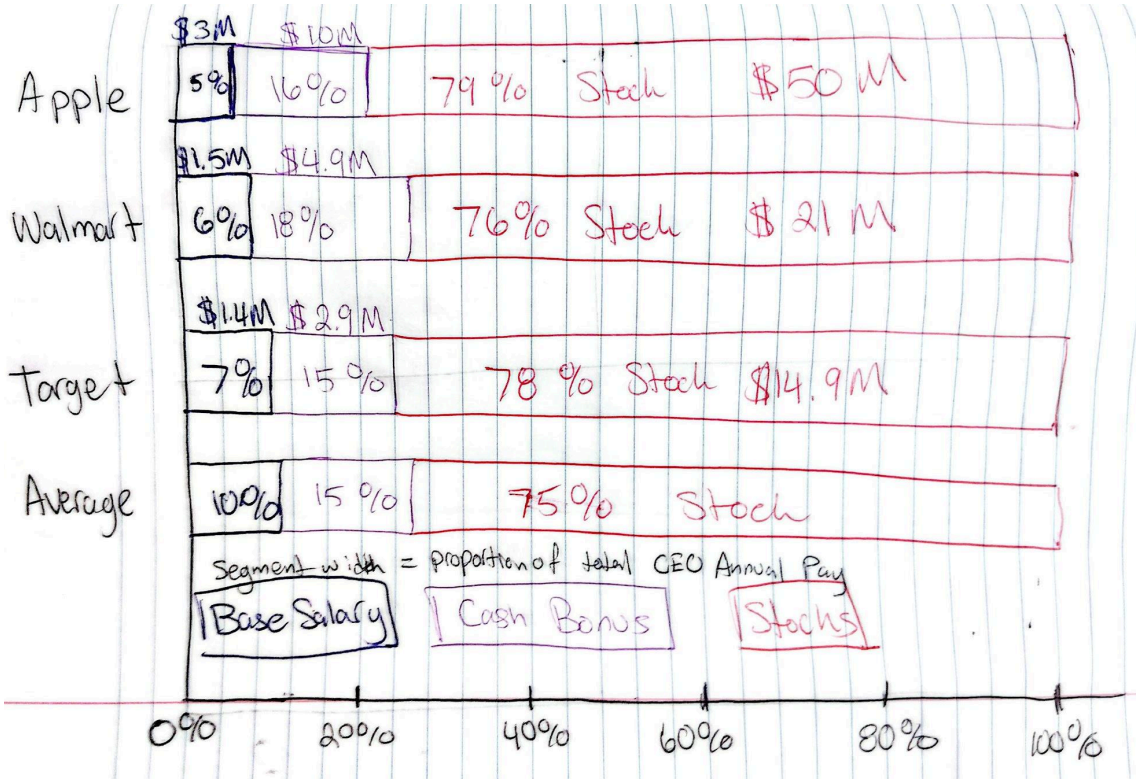
Which economic sectors (e.g., Retail/Service, Technology, Manufacturing) exhibit the highest pay disparities? Which exhibit the lowest?



How does CEO-to-worker pay disparity differ between unionized and non-unionized companies?

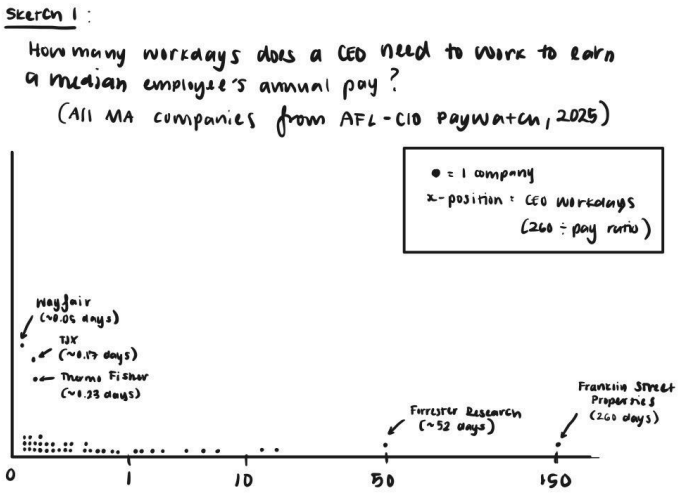


What proportion of CEO compensation comes from salary versus bonuses, stock awards, and options?

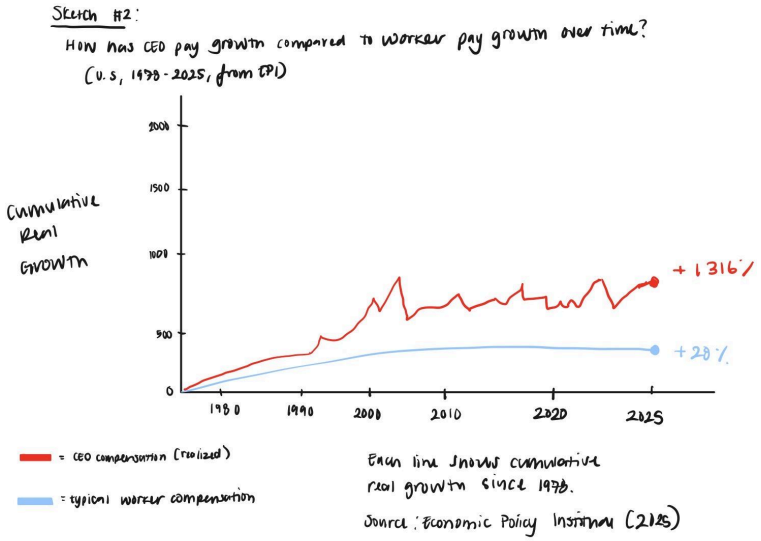


Kayleen Mukire

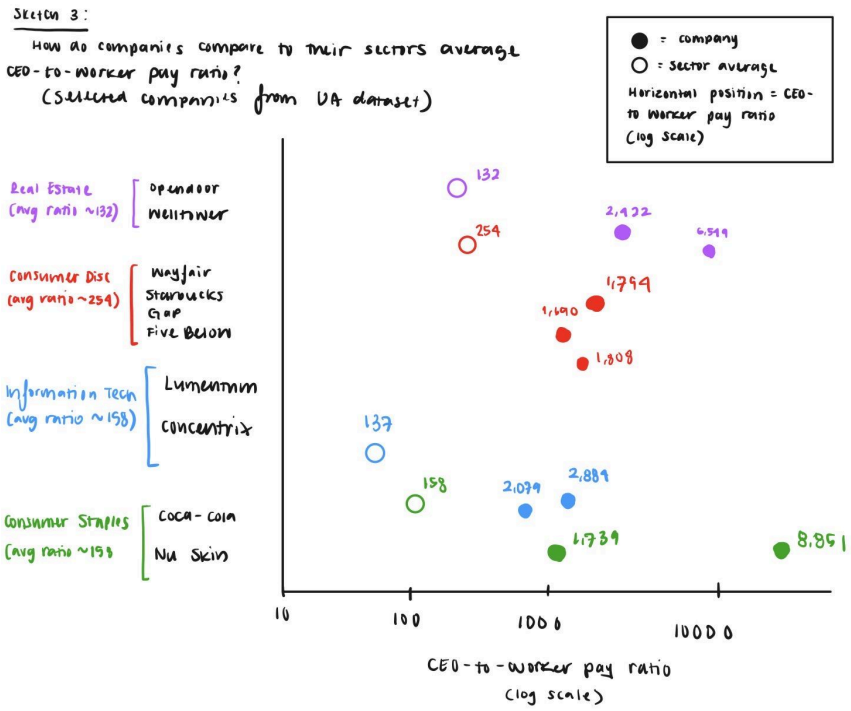
How many workdays does a CEO need to work to earn a median employee's annual pay?



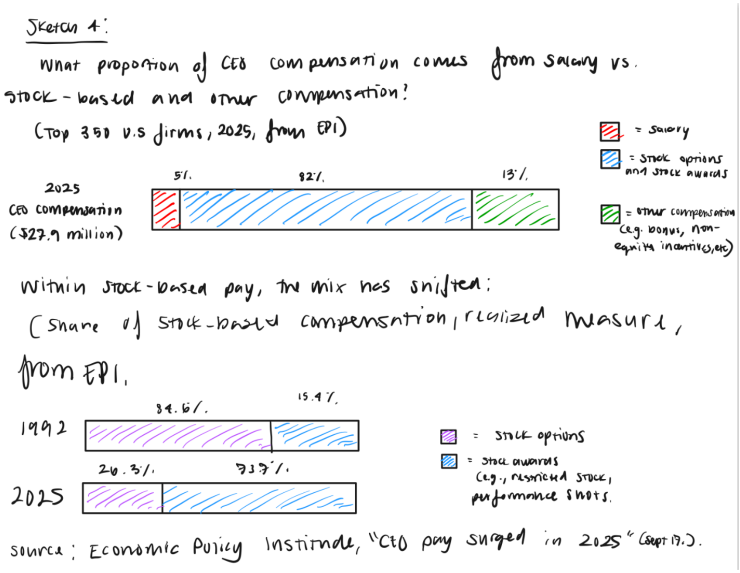
How does CEO pay growth compared to worker pay growth over time?



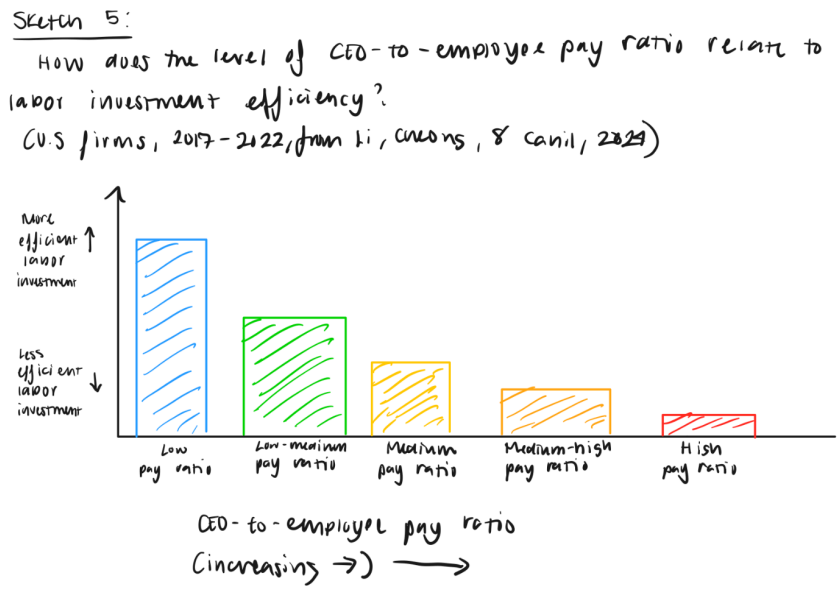
How do companies compare to their sector's average CEO-to-worker pay ratio?



What proportion of CEO compensation comes from salary vs. stock-based and other compensation?

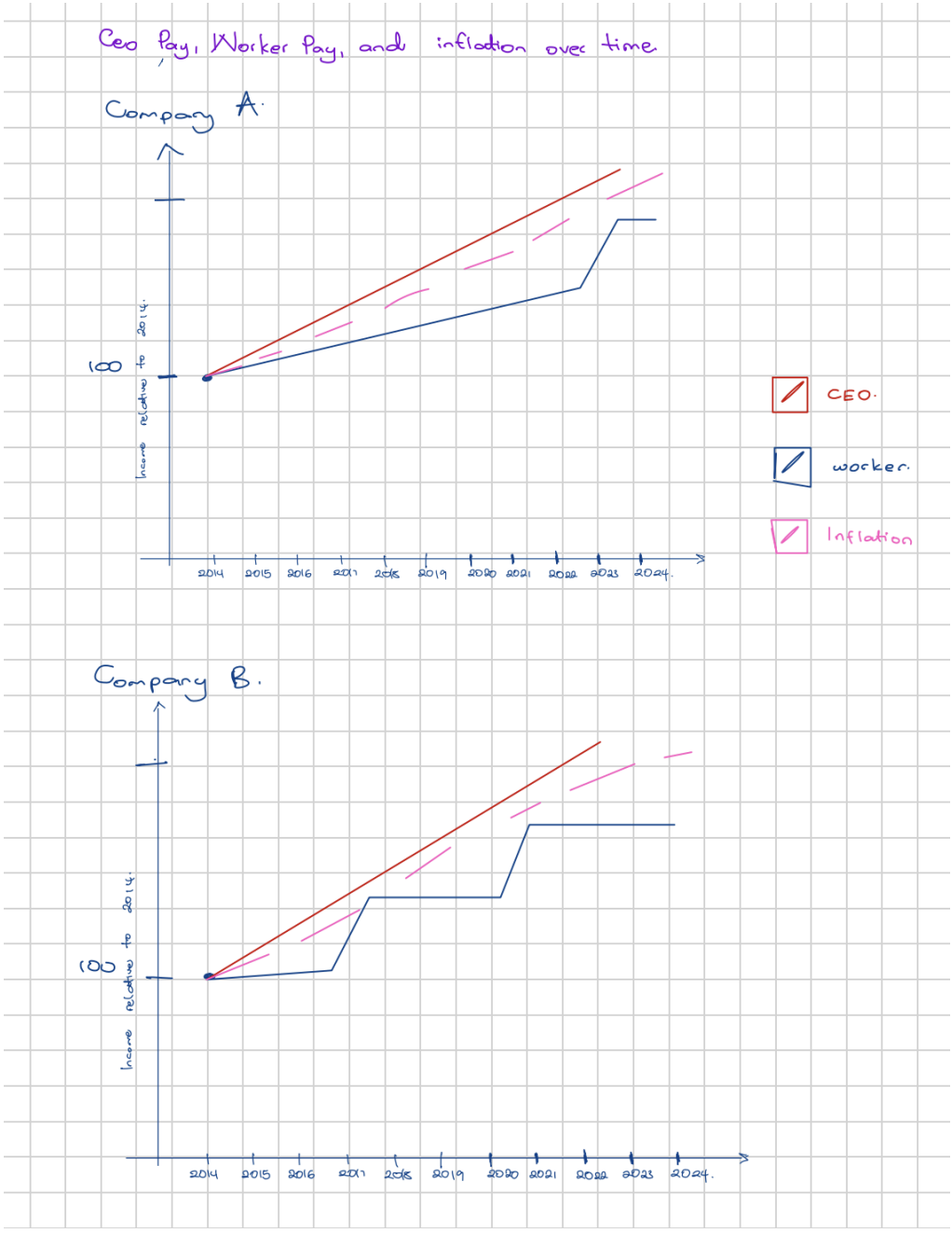


How does the level of CEO-to-employee pay ratio relate to labor investment efficiency?

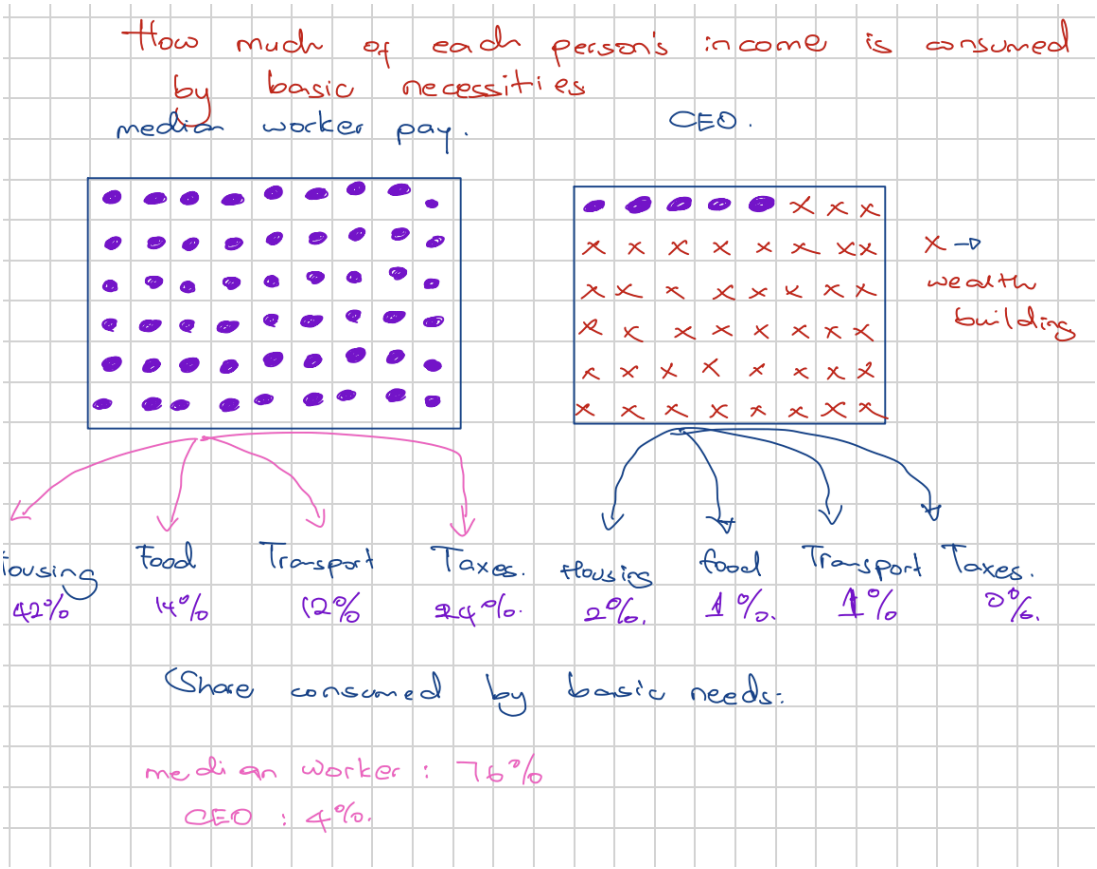


Aisha Ahmed

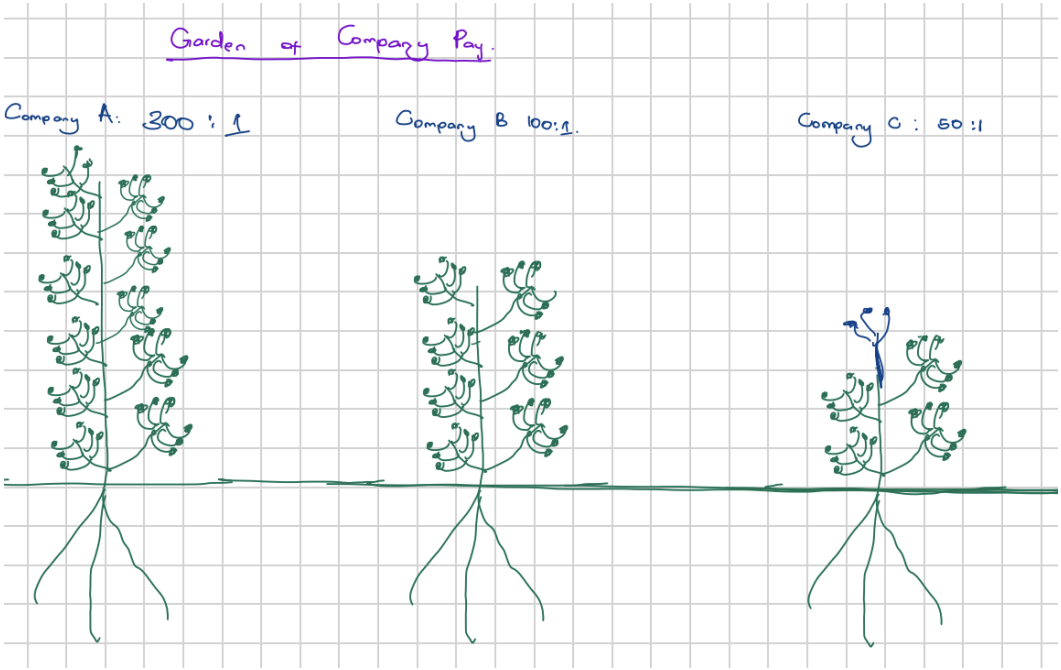
Question the Sketch is Answering #1



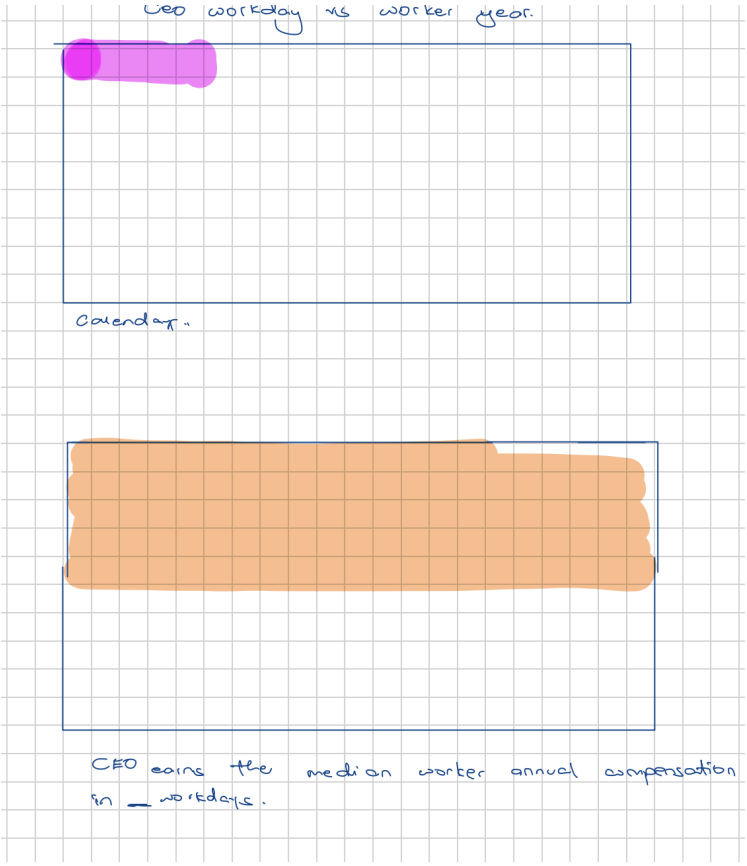
Question the Sketch is Answering #2



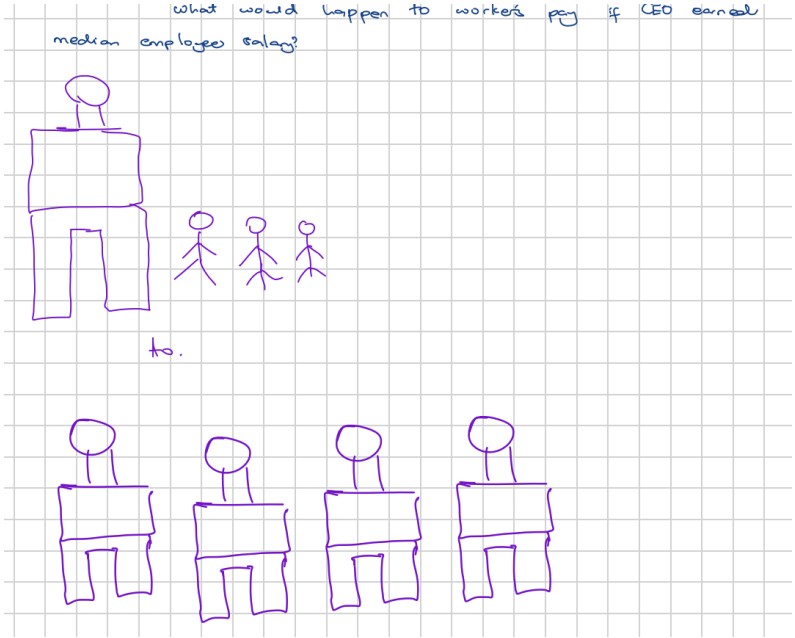
Question the Sketch is Answering #3



Question the Sketch is Answering #4



Question the Sketch is Answering #5



Milestone 4 | Decide (9/26/26)

Legend:

- Each sketch from Milestone 3, in order of appearance, will be labeled and referred to with an increasing Sketch ID from 1-15.
- Each question from Milestone 1: Interesting Questions For Our Audience (About Our Data), is already numbered, and will be used as the Question ID from 1-15
- Initials — Ashle Sanchez (AS⭐), Aisha Ahmed (AA🌟), Kayleen Mukire (KM✨)

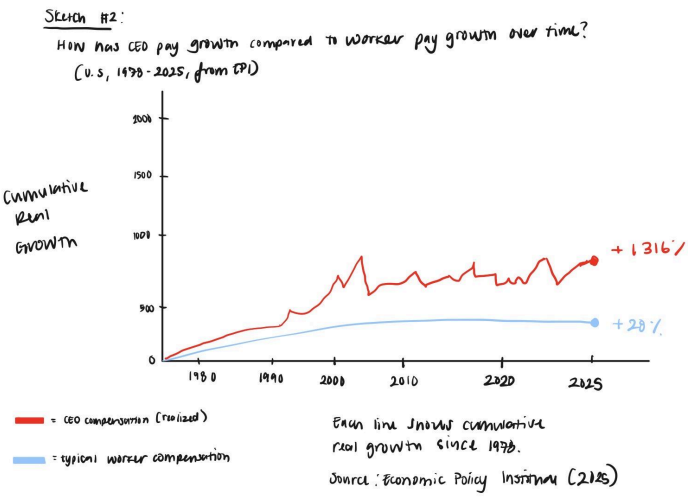
Sketch ID	Question ID	Author Initials	Total Votes (⭐🌟✨)
1, 6, 14	Q3, Q5 (unknowingly the exact same question)	AS, KM, AA	✨
2, 15	Q6	AS, AA	⭐🌟
3, 8	Q4	AS, KM	⭐✨🌟
4	Q8	AS	
5, 9	Q7	AS, KM	⭐🌟
7, 11	Q2, Q10	KM, AA	⭐🌟✨
10	Q9	KM	✨🌟
12	Q12	AA	⭐✨
13	Q1	AA	

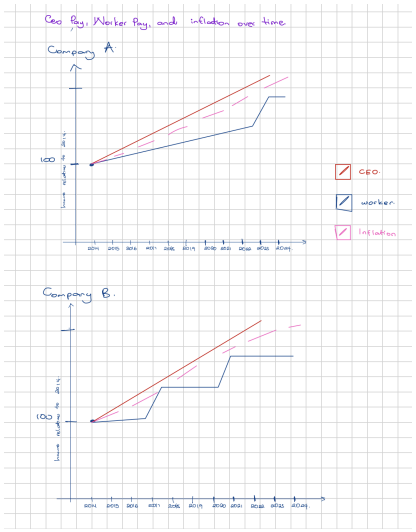
Notes

- Most relevant, good question with an interesting answer, can find a lot of data for several different companies across sectors (not a limited sample size) for question 4 (sketches 3 and 8)
- Audience would like to know more about whether pay has been keeping up with inflation and COL in comparison to the ever-growing CEO pay (interesting to find companies that increase their worker pay along with inflation and COLI vs companies that don't) We could also explore geographical data since we can see that the median COL changes with inflation differently with geography. (Question 2 and 10)
- Union audience could make a point about how worker lives can improve based on their increased share instead of giving bonuses to the CEO or taking a share of the total profit (Q6)
- Related to Q6, however, ultimately we decided that it was not as important or interesting to look into other than realizing majority of CEO pay is based on stocks rather than base pay and bonuses (Q7)
- For step 3, asked to elaborate on several sketches, for example one was sketch 10 and we discussed a bit on ways to make the sketch a little more intuitive to understand for someone who may not be entirely familiar with terminology or meaning of high pay ratio vs low pay ratio, and more efficient labor investment vs less efficient labor investment, and we agreed this was a really important graph to show (as long as we can improve the readability) as it showcases that companies can have great margins and efficient labor investment in comparison to corporations with a higher pay gap between CEO and median worker (Q9)
- Not much comparative data is available between different companies since all pay gaps look the same for median workers and CEOs. Still believe that showing an infographic (fixing the sketch) would be beneficial to audience as it would show workers living paycheck to paycheck with no extra money for leisure expenses or saving in comparison to executives using a fraction of their wealth with the rest for whatever else (can likely bring in examples of what they purchased) (Q12). We could also explore healthcare deserts/ utilities deserts since on average Median workers have realistically more costs in transportation due to geographical factors.

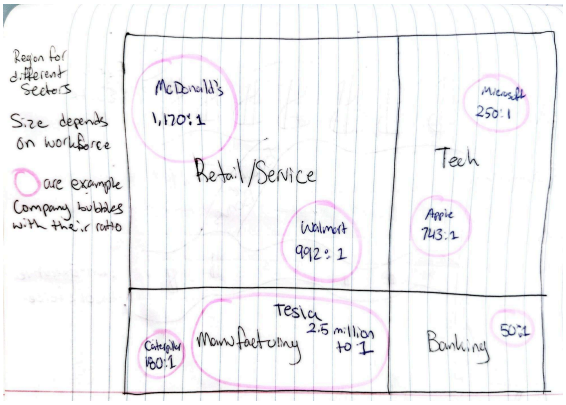
Post Voting Selected Sketches Ranking

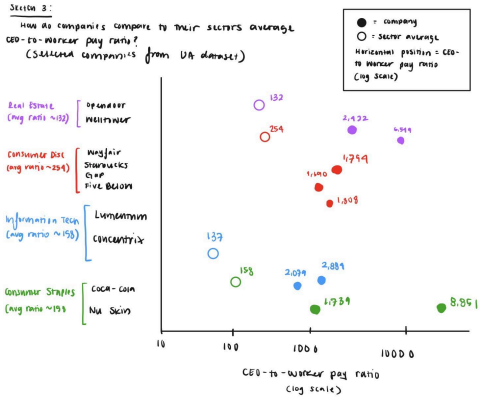
1) Questions 2 and 10, Sketches 7 and 11



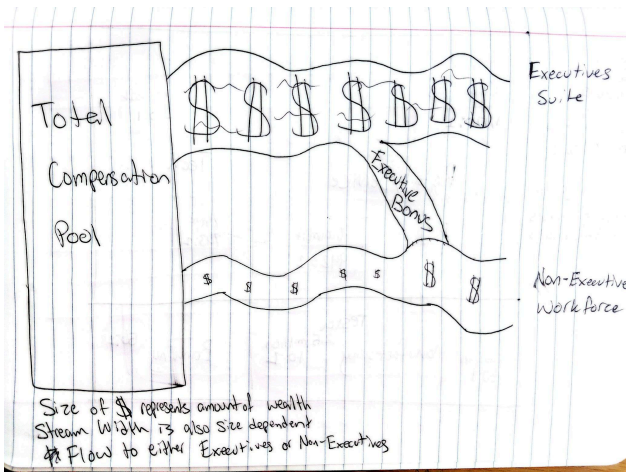


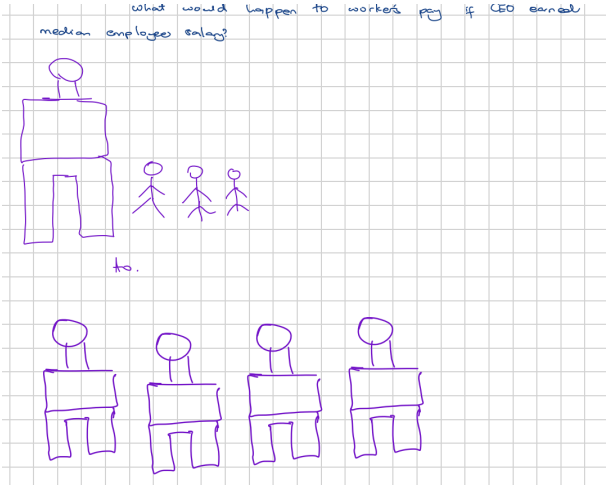
2) Question 4, Sketches 3 and 8



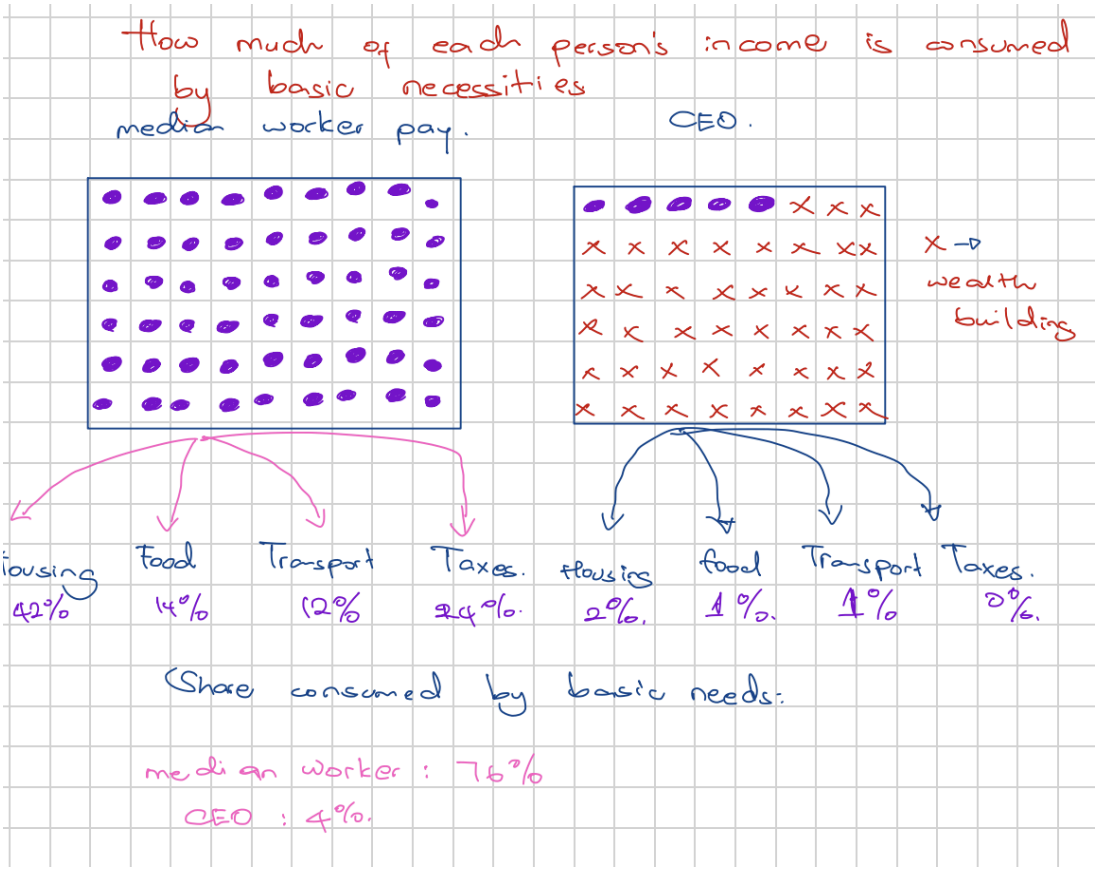


3) Question 6, Sketches 2 and 15

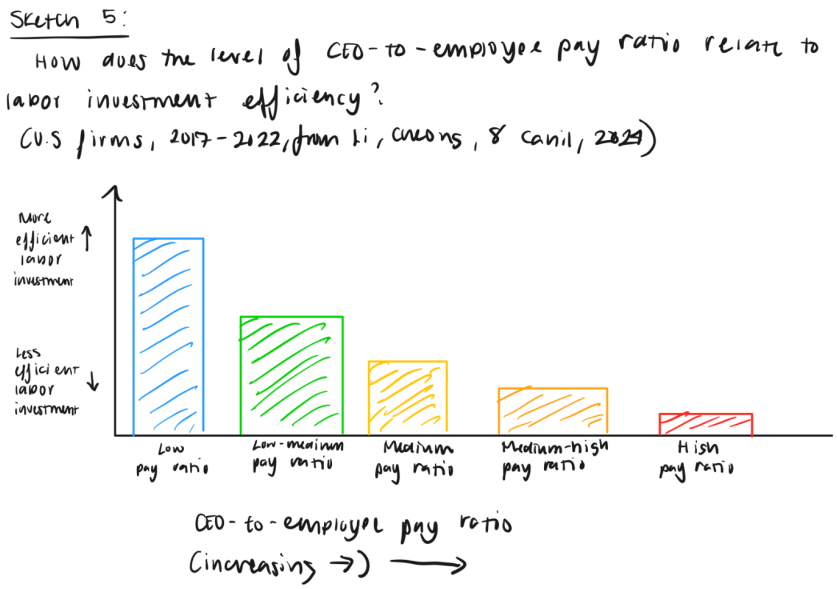




4) Question 12, Sketch 12



5) Question 9, Sketch 10



Summary

To follow the steps, we simply labeled each sketch with an increasing ID in order of appearance in Milestone 3, and same thing for questions (of which we added an additional based on a new sketch that was based off of an entirely new question). We created the table above, sorted by ID, and we combined similar sketches that answered the same question, or combined questions answered by a single sketch. When going through the sketches to prepare for voting, we discussed several sketches and possible ways for improvement, one such example was Sketch 10 answering Question 9, and while discussing, we had a hard time understanding it, we weren't quite sure what the labels meant or if it was an efficient visualization. After discussing a bit more on ways we can improve the readability of the visualization by deciding to alter the labels and format so that the design is readable to our main audience of Labor Union Representatives and the general workforce. Afterwards, we decided to vote individually using stars based on our preferences of which visualizations looked best and which questions were more meaningful to answer. Once we narrowed down the questions, we ranked them individually and convened, our rankings were not far off from each other, so we easily negotiated our ranking based on the most important questions we want to not only answer, but would be meaningful for our primary audience. We prioritized questions 2/10 (sketches 7/11) because our audience needs clear evidence about whether or not typical worker pay has kept pace with inflation and local cost of living over the last few years, in comparison to the evergrowing executive pay. We intend to find data from different geographic locations to account for differences in cost of living, and government data for information on annual inflation and typical cost of goods, and data on different corporations for their executive and median worker pay. We ranked question 4 next (sketches 3/8) since it would provide context to workers from different economic sectors about which have the highest pay gap vs those that compensate their workers a little more fairly. With data from several companies, our audience could choose to compare the differences and use it to demand better compensation. We ranked question 6 next (sketches 2/15) as it is one of the more direct answers (but not flashy) to how a redistribution of either executive pay/bonuses or company profits could directly impact the lives of all their employees due to the increase in salary. An honorable mention was question 7 (sketches 5/9), however we ultimately decided this was not as important of a question since it would be repeatedly describing executive pay as having a small base salary, slightly larger bonus, and the majority of their income being their growing stocks, and we believed there was limited things labor union representatives could do with that information in comparison to the rest of the things on our list. On the lower end of the spectrum, although they should still be considered as highly valued, we ranked question 9 and 12, as we agreed they did a great job in creating a story, as sketch 10, with a bit of fine tuning in its labels and visual design (mentioned before), does wonders in showcasing that lower pay gaps typically align with better operational efficiency, while question 12 (sketch 12) shows how median workers spend the majority of their income on basic necessities (living paycheck to paycheck) like housing and transportation, while these things are only a fraction of executive's income. In the end, we decided on 5 specific questions that we would spend a little more time and effort into making better visualizations for, something of quality that could actually be presented to our primary audience.

Milestone 5 | Design Review

Before Meeting

We had not yet prepared any sources and saw that this was one of the 4 core topics that we would be asked to discuss: topic and story, data sources, key questions being answered, and sketches we selected and why. This is why I (Ash), looked into data sources that may contain information useful for answering the questions we chose to answer in Milestone 4. This is what was found:

Q2: How does worker wage growth compare to executive pay over the last decade(s)?

https://fred.stlouisfed.org/series/CPIAUCSL
https://data.epi.org

- **Contains:** U.S. CEO-to-worker pay ratio (1965–2025), worker pay index, and monthly inflation (CPI)
- **Limitation:** EPI gives the ratio only, not CEO pay in dollars, and only national totals

Q4: Which economic sectors exhibit the highest pay disparities? Which exhibit the lowest? For each, how does a randomly (or specifically) selected company compare against their peer/rival companies?

https://aflcio.org/paywatch/company-pay-ratios
https://www.kaggle.com/datasets/salimwid/latest-top-3000-companies-ceo-salary-202223

- **Contains:** about 2,175 companies with respective sectors, CEO total pay, median worker pay, and pay ratio (fiscal year 2021)
- **Limitation:** each company picks its own "median employee" (sometimes part-time or non-U.S.), which can create extreme outliers

Q6: If a specifically chosen company redistributed their CEO's bonuses evenly across its non-executive workers, what percentage increase would the median employee receive? What if the CEO earned the median, and the remaining income was redistributed similarly?

https://aflcio.org/paywatch
https://www.sec.gov/Archives/edgar/data/104169/000119312526173673/wmt-20260423.htm

- **Contains:** CEO pay broken into salary, bonus, and stock, plus median worker pay and employee count, from SEC filings (5 companies, fiscal year 2025)
- **Limitation:** most CEO pay is stock rather than bonus, and the sample is small

Q9: How does the CEO-to-worker pay gap relate to a company's profitability or financial performance?
https://www.sciencedirect.com/science/article/pii/S1544612324007918?via%3Dihub

- No data. Li, Cheong & Canil (2024) discovered the reverse of what we claimed earlier: **higher pay ratios are actually associated with "less inefficient" (i.e. more efficient) labor investment**. We intend to drop Q9, unless we say something about the unfairness of the value created by workers and not being correctly compensated.

Q10: How much purchasing power does the median worker's salary provide compared with the CEO's compensation when adjusted for local cost of living?
https://apps.bea.gov/regional/downloadzip.htm

- **Contains:** cost-of-living index by state (U.S. average = 100), 2008–2024
- **Limitation:** it compares places, not years, and we will use each company's headquarters state

Q12: How much of each person's income is consumed by basic necessities compared to their CEO?
https://www.bls.gov/cex/tables/calendar-year/mean-item-share-average-standard-error.htm

- **Contains:** spending on housing, food, transportation, healthcare, and taxes by income level (2023)
- **Limitation:** the highest group is the top 20% of households, so the CEO side will be an estimate

During Meeting

I (Ash) prepared a script or guide of ideas on what to go over for the meeting to cover all key points in Milestone 5, and this is how we prepared:

Topic and Story

- **Topic:** The gap between CEO pay and median worker pay
- **Audience:** Labor union representatives with medium to high visualization experience; they know the gap is big but maybe not how big?
- **Story:** CEO pay has grown far faster than worker pay, worker pay has barely kept up with the cost of living, and the gap varies a lot by sector

Data

- Company-level data (AFL-CIO Paywatch via Kaggle, SEC filings) and national/regional data (EPI, FRED, BEA, BLS)
- Main caveat: each company defines its own "median employee," so we use sector medians and a log scale to handle outliers

Key Questions

- **Q2/Q10:** Has worker pay kept up with inflation and the cost of living while CEO pay grew?
- **Q4:** Which sectors have the largest and smallest pay gaps?
- **Q6:** What would redistributing CEO pay mean for workers?
- **Q12:** How much of their income do workers spend on basic needs compared to CEOs?

Sketches and Why

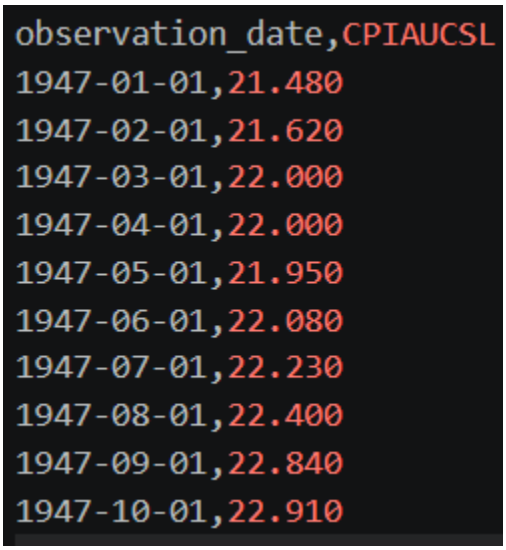
- **Q2/Q10 (Sketches 7, 11):** line charts over time provide the most direct evidence for our audience
- **Q4 (Sketches 3, 8):** sector comparison, so workers can compare their company with its peers/competitors.
- **Q6 (Sketches 2, 15):** redistribution of wealth: showcase direct impact on median workers' pay if it were increased accordingly to CEO pay or profits
- **Q12 (Sketch 12):** dot grid, turning dollar amounts into everyday life
- **Changes since Milestone 4:** dropped Q9 (research found the opposite); Q2 shows the pay ratio over time (EPI has no CEO pay in dollars); Q6 is reframed around the size of the gap or around profits/stock, since spreading the CEO's total pay adds only about $14 per Walmart worker.

During the meeting I (Ash) took notes on what was commented by the TF (Richard Morello) on ways to currently improve the design process up to this moment:

- Talk more about the data: what it contains and why?
- As we hone in on the questions, our audience could change
- What should our call to action be? (Consider the Dodd Frank Act)
- Why is certain data available and other data not?
- Create a Git environment for the project

Post Meeting

- **Data:** we documented what each dataset contains and its limitations in the Data Sources above, but to go over a few,
    - CPIAUCSL data contains a large csv file with two columns: dates (ranging back to January 1947 and monthly ever since) and CPI, which are both quantitative data



```
observation_date,CPIAUCSL
1947-01-01,21.480
1947-02-01,21.620
1947-03-01,22.000
1947-04-01,22.000
1947-05-01,21.950
1947-06-01,22.080
1947-07-01,22.230
1947-08-01,22.400
1947-09-01,22.840
1947-10-01,22.910
```

    - Kaggle data contains a lot of the bulk of our data as it has the median worker pay and pay ratio (along with CEO salary) from several different companies! In the following image, you can see several columns, and the ones that are categorical are: ticker which is the stock symbol, company name, ceo_name, and industry / sector, while quantitative columns include: median_worker_pay, salary, and pay_ratio



,ticker,company_name,median_worker_pay,pay_ratio,ceo_name,salary,industry
0,ATVI,"Activision Blizzard, Inc.","$117,207",7:1,Robert Kotick,"$826,549",Communication%20Services
1,CHTR,"Charter Communications, Inc.","$60,007",698:1,Thomas Rutledge,"$41,860,263",Communication%20Services
2,CMCSA,Comcast Corporation,"$83,840",405:1,Brian Roberts,"$33,978,581",Communication%20Services
3,DIS,The Walt Disney Company,"$50,430",644:1,Robert Chapek,"$32,464,293",Communication%20Services
4,DISCA,"Warner Bros. Discovery, Inc.","$82,964","2,972:1",David Zaslav,"$246,573,481",Communication%20Services
5,DISH,DISH Network Corporation,"$58,161",50:1,W. Carlson,"$2,895,882",Communication%20Services
6,EA,Electronic Arts Inc.,"$115,569",172:1,Andrew Wilson,"$19,858,539",Communication%20Services
7,FOXA,Fox Corporation,"$91,803",237:1,Lachlan Murdoch,"$21,748,681",Communication%20Services
8,GOOGL,Alphabet Inc.,"$295,884",21:1,Sundar Pichai,"$6,322,599",Communication%20Services
9,IPG,"The Interpublic Group of Companies, Inc.","$59,814",290:1,Philippe Krakowsky,"$17,371,647",Communication%20Services
10,LUMN,"Lumen Technologies, Inc.","$75,984",298:1,Jeffrey Storey,"$22,654,781",Communication%20Services
11,LYV,"Live Nation Entertainment, Inc.","$15,740",879:1,Michael Rapino,"$13,842,464",Communication%20Services
12,META,"Meta Platforms, Inc.","$292,785",92:1,Mark Zuckerberg,"$26,823,061",Communication%20Services
13,MTCH,"Match Group, Inc.","$106,881",238:1,Sharmistha Dubey,"$25,397,675",Communication%20Services
14,NFLX,"Netflix, Inc.","$201,743",202:1,Reed Hastings,"$40,823,725",Communication%20Services
15,NWSA,News Corporation,"$76,185",258:1,Robert Thomson,"$19,662,203",Communication%20Services
16,OMC,"Omnicom Group, Inc.","$50,612",395:1,John Wren,"$19,981,715",Communication%20Services
17,PARA,Paramount Global,"$94,580",212:1,Robert Bakish,"$20,035,212",Communication%20Services
18,T,AT&T Inc.,"$107,570",231:1,John Stankey,"$24,820,879",Communication%20Services
19,TMUS,"T-Mobile US, Inc.","$72,572",312:1,G. Sievert,"$22,610,659",Communication%20Services
20,TTWO,"Take-Two Interactive Software, Inc.","$70,096",2:1,Strauss Zelnick,"$142,996",Communication%20Services
21,VZ,Verizon Communications Inc.,"$122,492",166:1,Hans Vestberg,"$20,342,871",Communication%20Services
22,AAP,"Advance Auto Parts, Inc.","$24,960",403:1,Thomas Greco,"$10,052,271",Consumer%20Discretionary
23,AMZN,"Amazon.com, Inc.","$32,855","6,474:1",Andrew Jassy,"$212,701,169",Consumer%20Discretionary
24,APTV,Aptiv Plc,"$7,402","1,992:1",Kevin Clark,"$14,744,780",Consumer%20Discretionary
25,AZO,"AutoZone, Inc.","$31,751",451:1,William Rhodes III,"$14,330,010",Consumer%20Discretionary
26,BBWI,"Bath & Body Works, Inc.","$10,632","1,662:1",Andrew Meslow,"$17,668,627",Consumer%20Discretionary
27,BBY,"Best Buy Co., Inc.","$29,999",521:1,Corie Barry,"$15,631,157",Consumer%20Discretionary
28,BKNG,Booking Holdings Inc.,"$58,005",931:1,Glenn Fogel,"$53,982,195",Consumer%20Discretionary
29,BWA,BorgWarner Inc.,"$31,740",554:1,Frederic Lissalde,"$17,592,090",Consumer%20Discretionary
30,CCL,Carnival Corporation,"$8,658","1,740:1",Arnold Donald,"$15,063,788",Consumer%20Discretionary
31,CMG,"Chipotle Mexican Grill, Inc.","$15,811","1,131:1",Brian Niccol,"$17,880,580",Consumer%20Discretionary

- Another dataset worth talking about is the SARPP dataset, which contains data about the cost of living in several different states across the US from 2008 to 2024. Categorical columns include GeoFIPS (state id code), GeoName (state name), Region (numbered label in the US), TableName (referring to data csv—not important), LineCode (id for category in description), IndustryClassification (empty), Description (references to goods, housing, utilities, etc), Unit (referring to data csv—not important) while the years from 2008 to 2024 can be either ordinal or quantitative, but the values at each entry is quantitative



GeoFIPS,GeoName,Region,TableName,LineCode,IndustryClassification,Description,Unit,2008,2009,2010,2011,2012,2013,2014,2015,2016,2017,2018,2019,2020,2021,2022,
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"25000","Massachusetts",1,SARPP,1,"...","RPPs: All items ","Index",106.047,105.455,104.772,104.747,104.488,103.298,104.533,105.086,108.161,107.406,107.422,1

- **Audience:** we will revisit our audience constantly as we build our project and question whether the data actually supports and answers the questions we are trying to answer
- **Call to action:** After looking into the referenced Dodd-Frank Act, we found this required public companies to report their CEO-to-median-worker pay ratio. With more of this type of data available by law since 2017, we would want Labor Union Representatives to use the data for bargaining better wages and benefits.
- **Data availability:** After researching Dodd-Frank, this helped explain a lot of our data gaps, as company pay ratios only existed from the first report in 2018 and onward, and each company chooses its own method of reporting. This has come to be a huge hassle in trying to find relevant data on the topic of specific ratios, and long-term trends come only from EPI's national estimates, and pay data by worker or location is not reported.
- For our Git repository, it has already been made: https://github.com/aisharashidahmed/CS1710-Final_Project

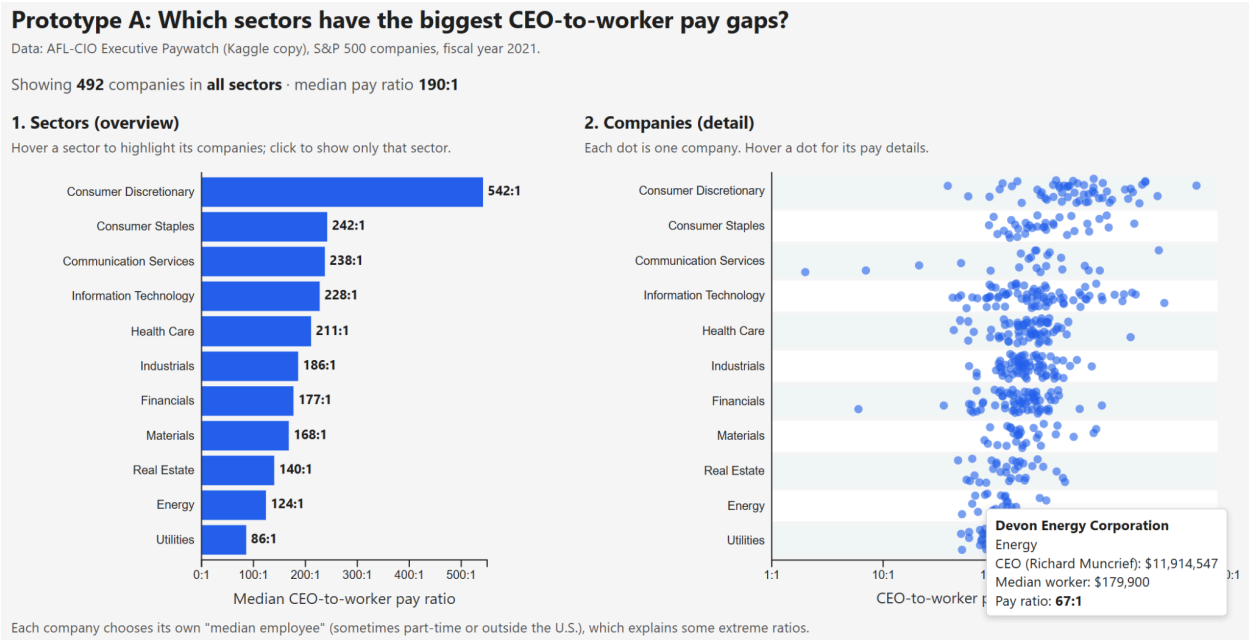
Milestone 6 | Interactive Dashboards

For Milestone 6, I (Ash) wanted to get some drafts that we may or may not actually use in the final deliverable, so these are ROUGH drafts and we are more likely to revisit and redo them when given more time to coordinate. The question that I chose to look into and answer with the Kaggle dataset I described in Milestone 5 is Q4 because it looked the most interesting to work with (also mainly because it was the only data source that I have actually really looked into), so I wanted the first drafts to focus on it. The Kaggle dataset compliments this question of: Which sectors and companies have the biggest gaps between CEO pay and median worker pay, because it directly contains fiscal 2021 data of the top 500 companies' (S&P 500) CEO salary, median worker pay, pay ratios, and the sector the company belongs to. Given previous sketches, we had Sketch 3 and 8 to work with, which mapped a company to their sector and then tried to magnify the scale of the pay ratio, and this premise was used in the prototypes. Together, they could let a Labor Union Representative find their respective sector and see where their own company falls within it. Another sketch that could be appended (or information to

include) is Sketch 1's idea of answering how many workdays a CEO needs in order to earn a median salary, and this could work well together with the other sketches by showing how big the gap is for a particular sector, and then look further into them by acknowledging the companies with the large pay gaps and provide the shock factor of "wow, they only need this many days to earn my salary?" Changes that were made from the sketches however were converting sectors to a sorted bar chart based on median pay ratio since comparing bar lengths was explained in lecture to work better at distinguishing lengths than areas and such.

Prototypes

Prototype A:
https://ash-harvard-le.github.io/CS1710_Final_Project_Personal/prototype-a/
Prototype B:
https://ash-harvard-le.github.io/CS1710_Final_Project_Personal/prototype-b/
Code: https://github.com/Ash-HARVARD-le/CS1710_Final_Project_Personal
Note: I couldn't actually host the prototypes using GitHub Pages since it requires the repo owner (Aisha) to change the settings, so I created my personal repo to showcase the prototypes for Milestone 6, but hopefully we intend to move most of our final work into the repo originally created in Milestone 5.





For prototype A, with the help of Claude, we created visualizations focusing on the sector's pay ratio and then showing individual companies in a specialized view with interactable features. The bar chart is sorted by highest to lowest median CEO-to-worker pay ratio for each of the 11 sectors listed in the data, while the dot chart has one dot per company, placed by pay ratio (log scale for spacing purposes but this could be harder to read/understand) in its sector's row. Similar to the lab, if we hover over a sector bar, then that sector's companies stay highlighted while those from others fade / get a lower opacity. If the user were to decide to click a sector bar, then the dot chart updates accordingly by only showing the sector, and updating the information labeled above such as the # of companies showed, which sector, and the bolded pay ratio. There is a button to reset everything, but the user can also just click the bar chart again to undo the filtering. Finally, hovering over a dot shows the company that dot represents, along with info about their CEO, their pay, median worker pay, and pay ratio.

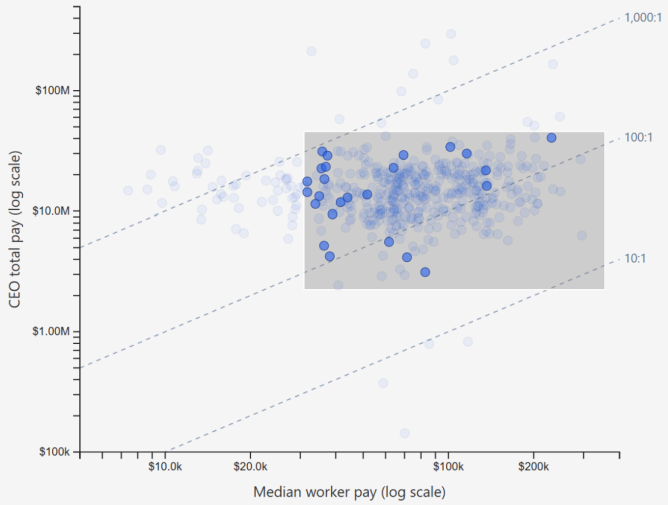
Prototype B: How do CEO pay and worker pay compare, company by company?

Data: AFL-CIO Executive Paywatch (Kaggle copy), S&P 500 companies, fiscal year 2021.

426 of 492 companies selected · median pay ratio **176:1**

1. Every company

Drag on empty space to select companies; click empty space to clear. Click a dot for its details. Dashed lines mark equal pay ratios.

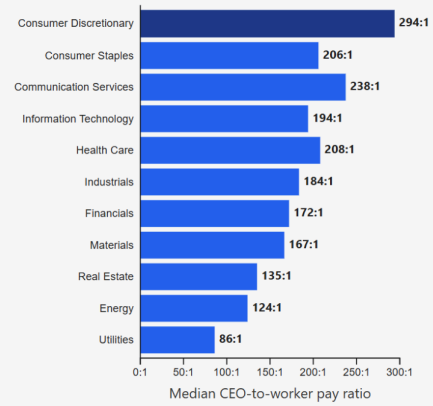


2. One company: the CEO's workdays

Click a dot to see how long its CEO takes to earn a worker's yearly pay.

3. Selected companies by sector

Hover a sector to highlight its companies in the chart.



Workdays assume 260 workdays per year. Each company chooses its own "median employee" (sometimes part-time or outside the U.S.).

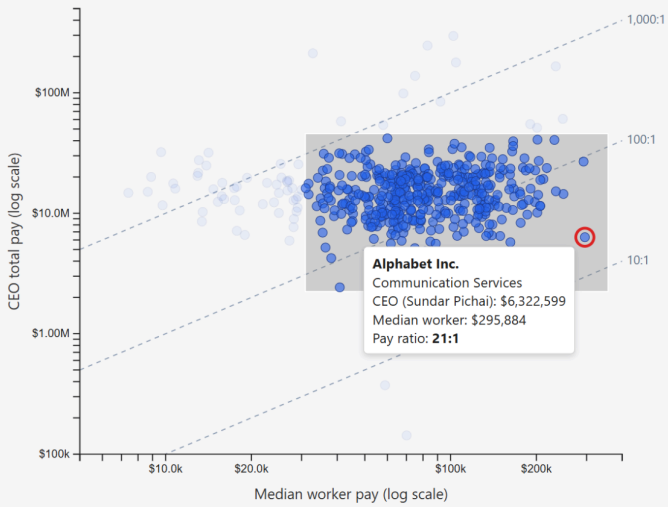
Prototype B: How do CEO pay and worker pay compare, company by company?

Data: AFL-CIO Executive Paywatch (Kaggle copy), S&P 500 companies, fiscal year 2021.

426 of 492 companies selected · median pay ratio **176:1**

1. Every company

Drag on empty space to select companies; click empty space to clear. Click a dot for its details. Dashed lines mark equal pay ratios.



2. One company: the CEO's workdays

Alphabet Inc.

12.4 workdays

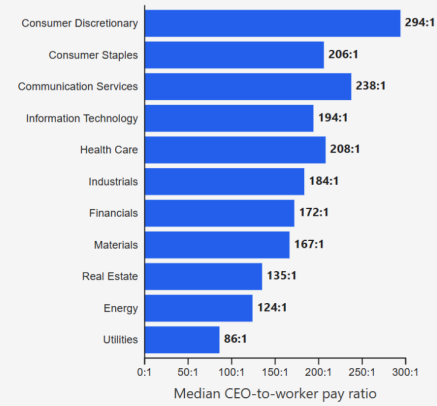
is how long it took CEO Sundar Pichai to earn what the company's median worker earns in a full year ($295,884).

Jan 1 — Dec 31 (worker's full year)

Pay ratio 21:1: the median worker would need about **21 years** to earn the CEO's pay for one year.

3. Selected companies by sector

Hover a sector to highlight its companies in the chart.



Workdays assume 260 workdays per year. Each company chooses its own "median employee" (sometimes part-time or outside the U.S.).

For prototype B, we created a company scatterplot, with bonus information about the company and how long it would take for a CEO to earn their median worker's salary, and the sectors like in prototype A. The scatterplot shows every company with the median worker pay on the x-axis and the CEO total pay on the y-axis, on log scales (again, due to spacing issues, but I want to try to get it to work so it is a potential future improvement) and dashed lines to reference ratios like 10:1, 100:1, and 1,000:1. Once a company is clicked on, we can see specific company information in the top right white space that fills in with the company name, CEO name, and the time it takes for them to earn their median worker's salary with a progress bar corresponding to how far into the year it is. Then the bar chart sorts sectors by median pay ratio, and hovering over them highlights the corresponding dots on the scatterplot. Brushing and filtering is possible on the scatter plot, with the sector bars and summary line recomputing for the selected companies in the brush. The user can just click on the scatterplot again to reset.

Comparison

Personally, I like Prototype B because of its nicer layout and the information it provides, but Prototype A offers good features too. A is far simpler than B, but it quickly answers our main question of which sectors are worst in terms of pay gap right away because of the immediately apparent sorted bar chart and labeled median pay ratios. The interactions are easy to discover, since there are only bars and dots which a little bit of exploring can find you can hover and click on them. The biggest downside however is definitely the log scale in the dot plot, as looking at it I would assume a lot of these dots are close together, but in reality, because of the log scale, this is NOT true (most clear example is Communication Services where the far left dot and the 3rd dot that follows are exceptionally far from each other, but have a differing pay ratio of 2:1 and 50:1, while the last two dots are closer but differ 879:1 and 2,972:1). B still shows the same bar chart, but requires a bit more effort to find, and emphasizes more on the scatterplot, directly emphasizing the relationship between CEO pay and median worker pay (still on on a log scale which can make things very difficult to understand), and shows dots closer together than in actuality. While there are nice reference lines to show the pay ratios, I would argue they barely help the plot since the biggest issue is the

clumped together dots at the center. One good thing about this plot is that clicking on dots allows you to see more in-depth information about a particular company, and you can find that it takes Amazon 0.3 work hours to make the median worker's annual salary! You can also brush to filter on the scatterplot, however, that may not be entirely clear to the user on what it means, nor would a box within a region be entirely helpful (for example, consider being able to brush from pay ratios 100:1 and 1,000:1 instead of a box centered at the clutter).
- Coordinating the views helps in both. In A, filtering to one sector makes it clear what you're looking at and puts more attention to the outliers in each sector, while in B, brushing scatterplot outliers is useful to see which sectors they come from.

Claude

Claude was the main contributor to the actual code, and checking was done by me (Ash). I found the data in Milestone 5 and downloaded it, and used it here in Milestone 6, where Claude was able to inspect it and use it to generate the SvelteKit and D3 code for both prototypes based on the interactive dashboard lessons from Lab 4. I was the one who chose the question based on the available data, which sketches to combine and what designs / interactability features should be present (highlighting, filtering, brushing, and hovering) as well as what plots should be visible. I tested both prototypes locally before pushing to ensure all features I prompted for were present. For the moment, I asked to use log scales for spacing issues on outliers, but hopefully fixable in the future, corrected the naming conventions Claude used based on the data to what they actually meant (like salary in the data but it meant CEO pay), and finally cherry picked random dots to check if the values computed on the data (mainly the workdays and workhours prototype B) were correct.

Milestone 7 | Storyboard

Individual Insights

For this milestone, we (Aisha and Kayleen) explored the fiscal 2021 S&P 500 dataset from Milestone 6 (492 companies with CEO total pay, median worker pay, pay ratio, and sector), using Python and the Milestone 6 prototypes.

Aisha Ahmed

Insight 1: Some median workers would need to have started working before the pyramids were built to earn their CEO's one-year pay. A company's pay ratio also tells us how many years its median worker would need to work to earn what the CEO made in a single year. At Amazon, CEO Andrew Jassy earned $212,701,169 in 2021 while the median worker earned $32,855, a ratio of 6,474 to 1. A median Amazon worker would have needed to start working in 4453 BCE, nearly 2,000 years before the Great Pyramid of Giza was built, to match one year of Jassy's pay. Seven companies push their worker's start date before year 1, including Western Digital (around 1311 BCE, during ancient Egypt's New Kingdom), Expedia (876 BCE, before Rome was founded), and McDonald's (230 BCE, during the Roman Republic). Across all 492 companies, 180 would require their median worker to have started before 1776, and the typical (median) company's worker would have needed to start in 1831.

Insight 2: At 34 S&P 500 companies, the median worker earns less than the federal poverty line for a family of four, and 31 of those companies paid their CEO more than $10 million. We compared each company's median worker pay with the 2021 federal poverty guideline for a family of four ($26,500). At 34 companies, the median employee earned less than that, while those companies' CEOs earned a median of $15.9 million. Over half of these companies (19 of 34) are in Consumer Discretionary, which includes restaurants, retail, and travel, where much of the workforce is part-time or seasonal.

Kayleen Mukire

Insight 1: The companies with the most extreme pay ratios are often not the ones with the highest-paid CEOs. When we ranked all 492 companies by CEO pay and separately by pay ratio, the two rankings were only moderately related (Spearman correlation of 0.59). McDonald's ranks 125th in CEO pay but 5th in pay ratio, because its median worker earned just

8, 897. *Aptivranks*241*stinCEOpaybut*8*thinratio, andCarnivalranks*232*ndinCEOpaybut*11*thinratio. T*
40.6 million), but Etsy ranks 277th in pay ratio because its median worker earned $231,338. So a high ratio can come from very high pay at the top, very low pay at the bottom, or both.

Insight 2: Companies in the same sector differ far more from each other than sectors differ from one another. Sector medians range from 86:1 (Utilities) to 542:1 (Consumer Discretionary), but within Consumer Discretionary alone, companies range from 38:1 to 6,474:1. Information Technology ranges from 42:1 to 3,332:1, and Communication Services from 2:1 to 2,972:1. (The lowest ratios mostly come from CEOs who take very small salaries but hold large amounts of company stock, such as Take-Two Interactive and Berkshire Hathaway, which we plan to note in the final visualization.) Industry alone does not determine how large the gap is.

Team Discussion

When we reviewed our insights together, we noticed that simply showing a large ratio would not tell our audience anything new. Labor union representatives already know the gap is big. We decided to focus on three insights that make the gap *felt*, *surprising*, and *personal*.

The first is Aisha's pyramid timeline. Saying "6,474 to 1" is abstract, but saying a worker would have needed to start before the pyramids is memorable and easy to retell, which matters for an audience that campaigns and organizes.

The second is Kayleen's ranking comparison. It shows that the companies with the worst ratios are not always the ones with famous, highly paid CEOs; companies like McDonald's reach extreme ratios mainly through very low worker pay. This helps union representatives identify whether the problem at their company is excessive pay at the top or insufficient pay at the bottom, which calls for different bargaining strategies.

The third is a personalized "pay receipt." Across all 492 companies, the typical CEO earns their median worker's annual pay in about 10.9 work hours, before lunch on their second day of work. At Amazon, it takes about 19 minutes. We plan to let readers choose their company and receive a shareable receipt showing when

their CEO "earned their salary" and what year they would have needed to start working to match their CEO. This turns our data into campaign material union representatives can share directly.

Aisha's poverty-line finding and Kayleen's within-sector finding will support the story as rising insights rather than being dropped. The within-sector spread in particular forms the basis of our main message.

Main Message

"A single year of CEO pay can outlast all of recorded history in worker wages, and because the gap varies so widely even between competitors in the same industry, it is a choice that workers can challenge using the pay data companies are now required to disclose."

We chose this message because it pairs the emotional impact of the pyramid timeline with a takeaway our audience can act on. Since the Dodd-Frank Act's disclosure rule took effect in 2018, public companies must report their CEO-to-median-worker pay ratio, so union representatives already have access to this data. If every company in an industry had the same gap, it might seem unavoidable. Because Consumer Discretionary companies alone range from 38:1 to 6,474:1, representatives can point to their competitors as proof that a smaller gap is possible.

Storyboard

Hook: The page opens with a live counter. Amazon's CEO earned about $6.74 per second in 2021, while the median Amazon worker earned about a tenth of a cent per second. In five minutes of reading, the CEO "earns" about $2,000 and the worker about 31 cents. Readers are then invited to pick their company.

Rising Insights: (1) The pyramid timeline places each company's median worker on a historical timeline of when they would have needed to start working, from Amazon in 4453 BCE to the median company in 1831. (2) A sorted sector bar chart, followed by a dot plot showing how widely companies vary within each sector. (3) A slope chart showing how company rankings change between CEO pay and pay

ratio, highlighting McDonald's, Aptiv, Carnival, and Etsy. (4) A callout showing the 34 companies whose median worker earns below the poverty line.

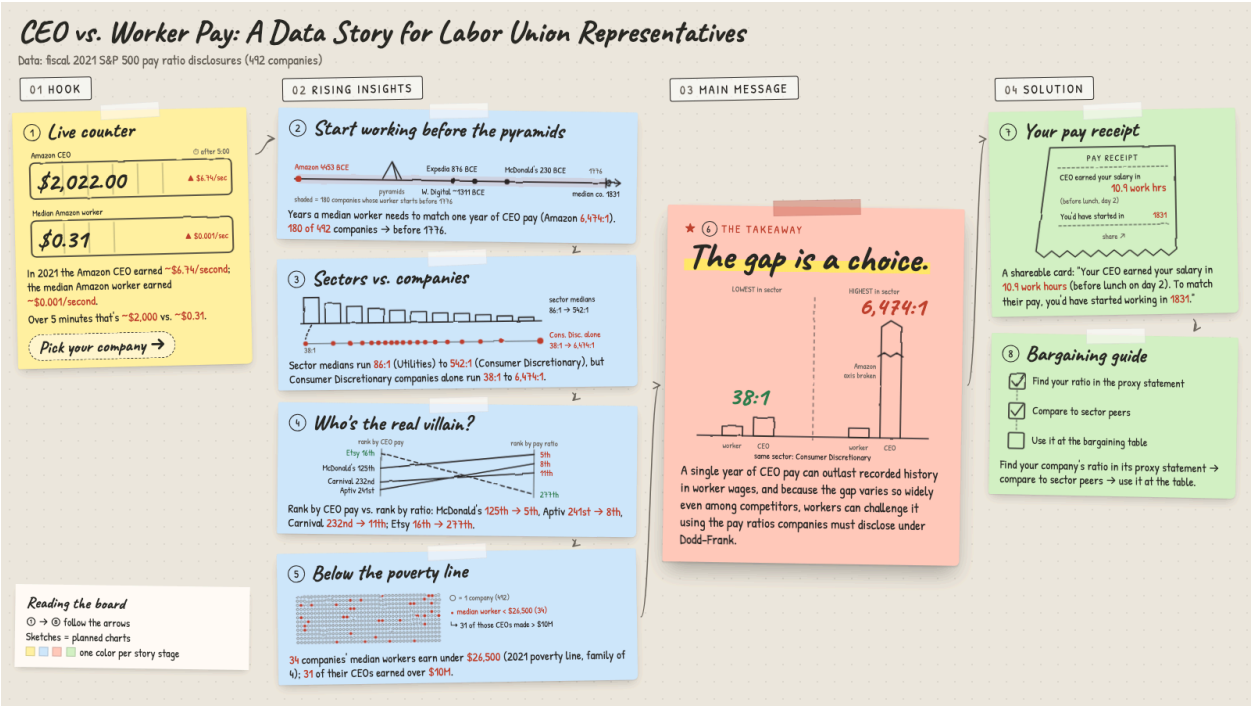
Main Message: The most and least equal companies in each sector are shown side by side, making clear that the size of the gap is a company decision rather than an industry rule.

Solution: The personalized pay receipt, downloadable and shareable, plus a short guide for union representatives: find your company's ratio in its proxy statement, compare it to sector peers, and use those comparisons in bargaining.

Claude

Claude helped compute the statistics above from our Milestone 6 dataset and helped draft the written descriptions, which we reviewed and edited. We chose which insights to pursue, selected the three for our story, decided on the main message, and designed the storyboard structure. Claude was also used to generate the digital storyboard image from our plan.

Screenshot for the storyboard:



Milestone 8 | Prototype I

Live link: https://aisharashidahmed.github.io/CS1710-Final_Project/
Code:
https://github.com/aisharashidahmed/CS1710-Final_Project/tree/main/static/data

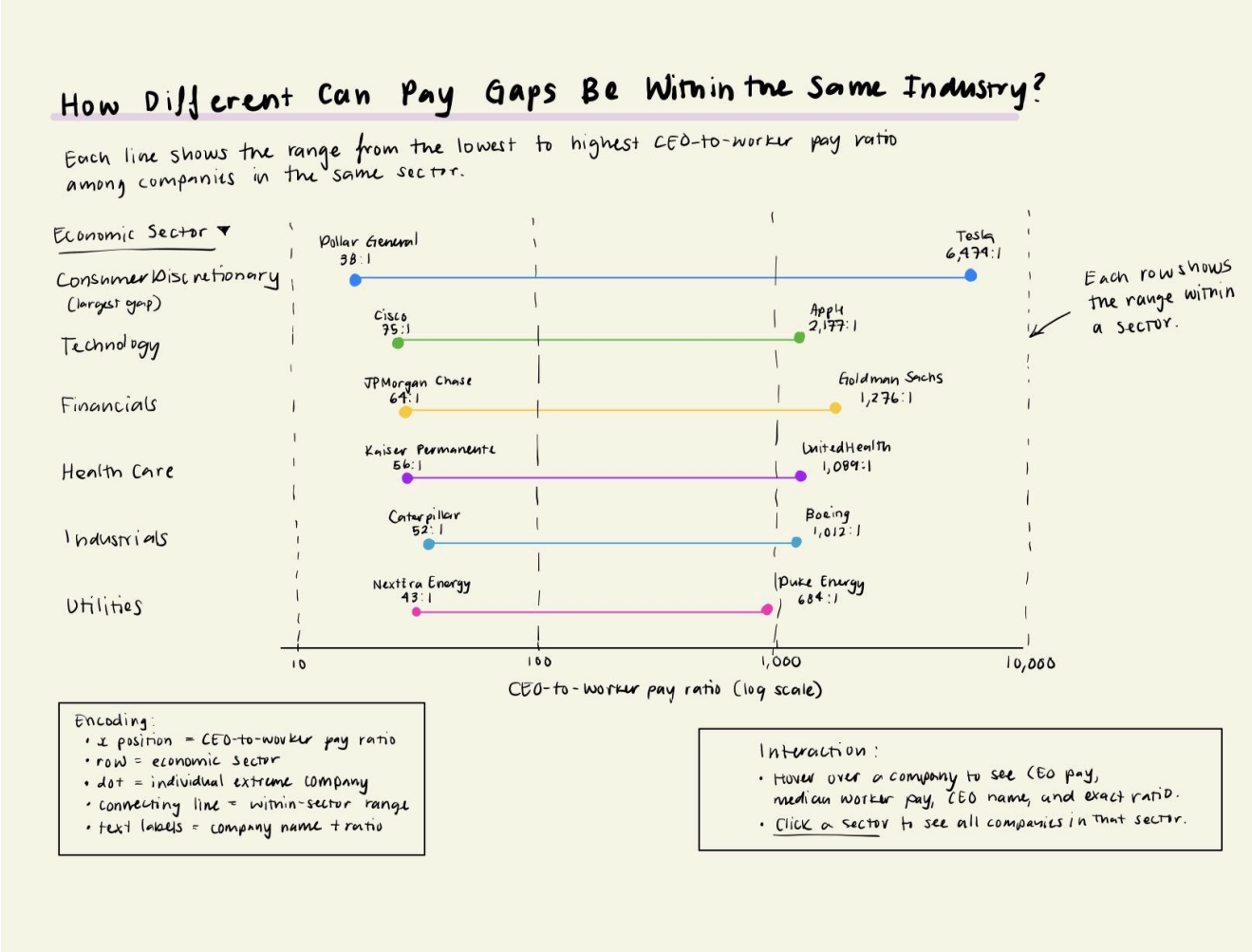
Data Pipeline

- **Raw data:** AFL-CIO Executive Paywatch (Kaggle copy), 492 S&P 500 companies, fiscal year 2021.
- **Cleaning (npm run clean):** text money and ratios turned into numbers, sector names fixed, "salary" renamed to CEO total pay, rows with missing values dropped (none).
- **Added columns:** workdays and work hours for the CEO to earn the median worker's yearly pay, start year (2021 − ratio), below the poverty line ($26,500, family of four), and CEO pay rank and pay ratio rank.

List of potential chart designs to sketch

Placeholder	Shows	Interaction
Pyramid Timeline	5000 BCE → today, one mark per company at its worker's start year, with the Great Pyramid and 1776 as landmarks	hover a mark, search for your company
Slope Chart	CEO pay rank vs. pay ratio rank, labeling McDonald's, Aptiv, Carnival and Etsy	hover a line to highlight it
Poverty Callout	the 34 companies under $26,500, CEO pay next to worker pay	filter by sector
Most vs. Least Equal	the lowest- and highest-ratio company in each sector	pick a sector
Pay Receipt (novel view)	a receipt for one company: both salaries, when on Jan 1 the CEO had already earned a worker's year, and the worker's start year	search for a company, download or share

Additional Visualization Draft

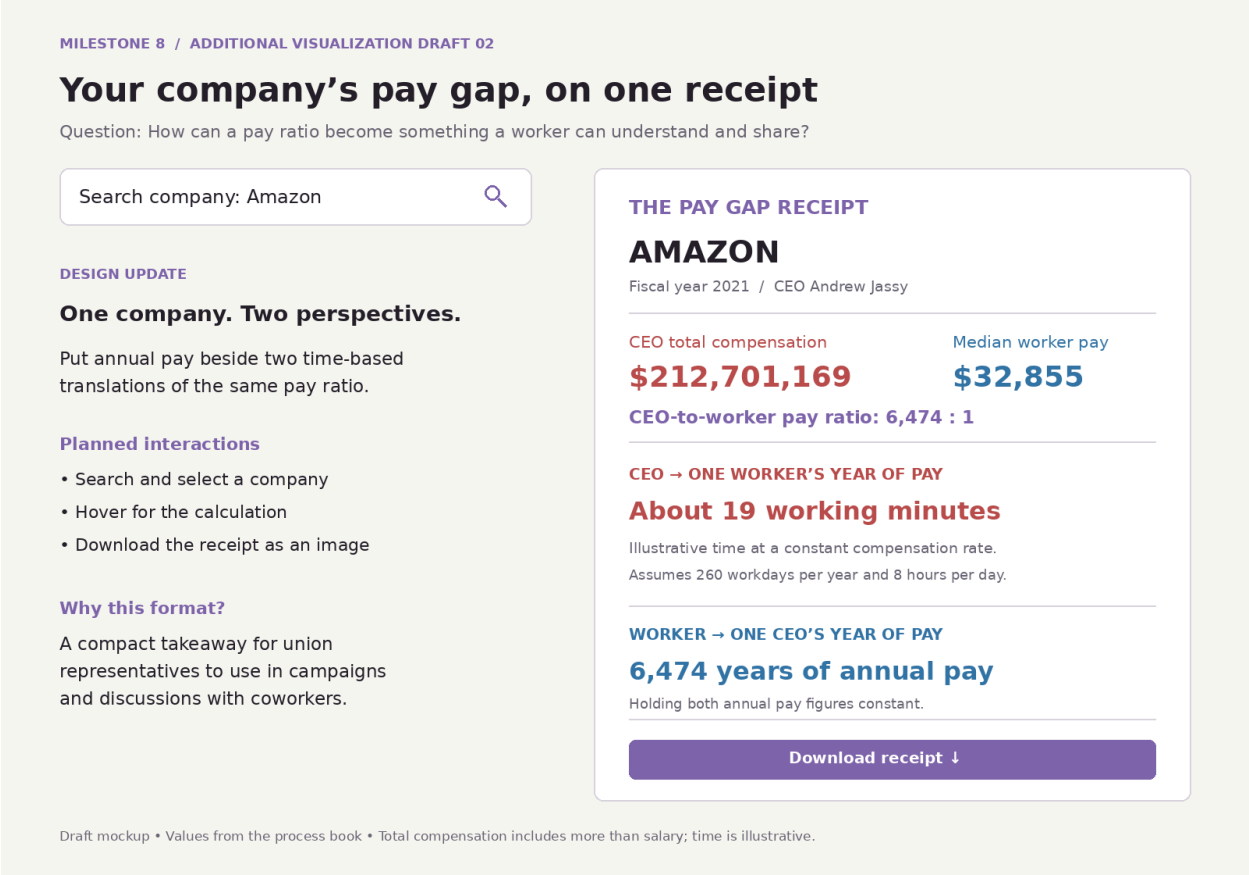


Question: How much can CEO-to-worker pay disparity vary between companies in the same economic sector?

Kayleen designed this visualization as a horizontal range plot with one row for each sector. The left point represents the company with the lowest CEO-to-worker pay ratio in that sector, while the right point represents the company with the highest ratio. Horizontal position encodes the pay ratio, and the connecting line emphasizes the spread within each sector. Kayleen chose this design because it directly supports the team's main message that companies in the same industry can still have very different pay gaps. A future interaction could let readers hover

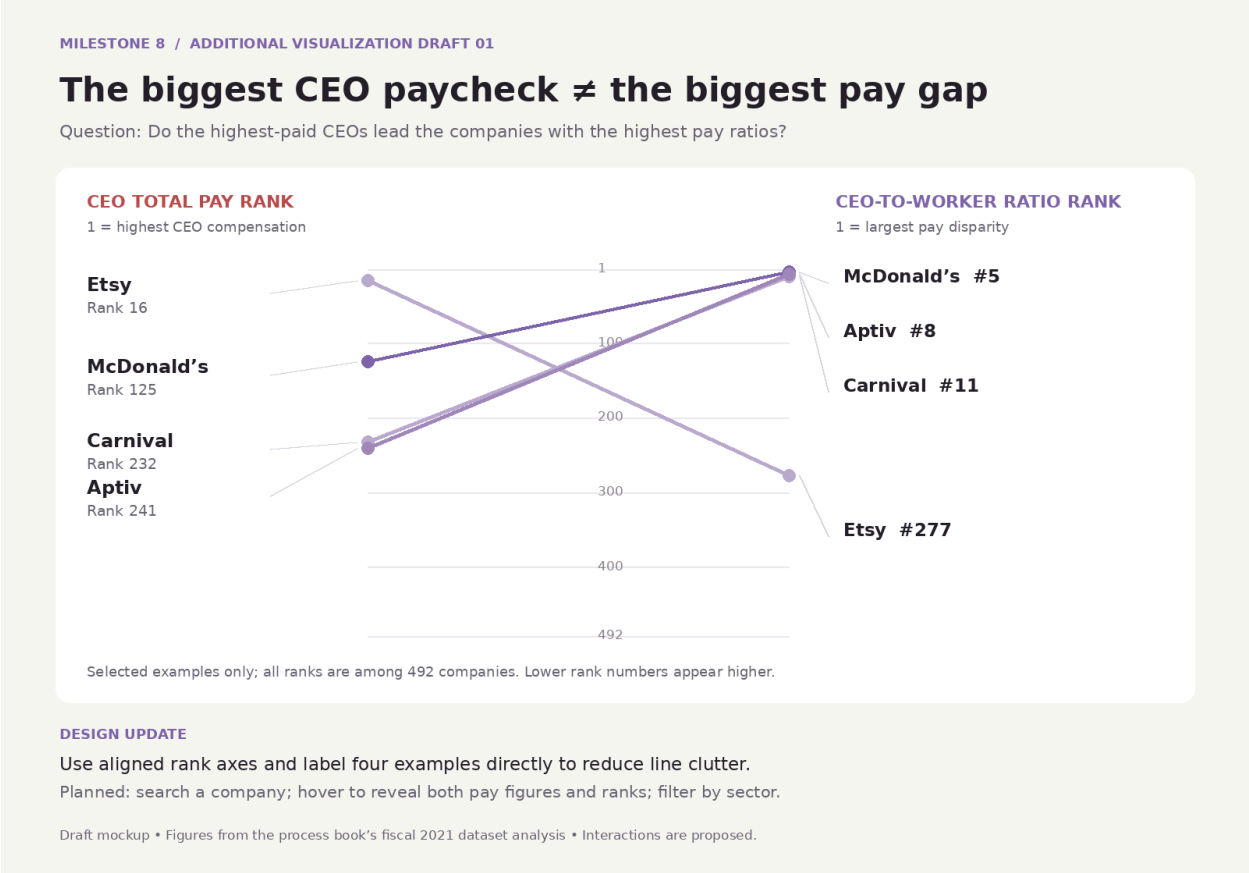
over a company to see CEO pay, median worker pay, and the exact ratio, or click a sector to view all companies in that industry.

Aisha's Visualization Drafts:



Question: How can we make a company's pay gap personally meaningful and easy to share?

This updated design presents one company's pay figures as a receipt, using red for CEO compensation, blue for worker pay, and purple for the company's ratio. It translates the ratio into two perspectives: the working time a CEO would need to match a worker's annual pay, and the years a worker would need to match the CEO's annual compensation. Compared with the storyboard's January 1 timestamp, this version states the working-time assumptions directly. Readers could search for their company, inspect the calculations, and download the receipt for organizing or discussions with coworkers.



Question: Are the companies with the highest-paid CEOs also those with the largest CEO-to-worker pay gaps?

This updated design uses a slope chart connecting each company's CEO pay rank to its pay-ratio rank. Both axes use the same scale, with rank 1 at the top. Rather than displaying all 492 companies at once, the draft labels McDonald's, Aptiv, Carnival, and Etsy to make their contrasting rankings easier to follow. Planned interactions include searching for a company, filtering by sector, and hovering to reveal both pay figures and rankings. This comparison helps union representatives distinguish high CEO compensation from low median worker pay as contributors to a large gap.

How we used color: We use a consistent color story across the site: red always means CEO pay, blue always means worker pay, light purple marks a company (which has both), and gray marks a sector, so union representatives can read every chart without a legend.

How we used Claude:

- Claude wrote the code commit by commit (SvelteKit + D3), reusing the two Milestone 6 prototypes as the working charts
-
- Ash and Aisha decided the page structure from our Milestone 7 storyboard, which charts to build now versus later, and which interactions to keep
    - For example, we dropped the scatterplot brush because in Milestone 6, Ash said it was too confusing and added no value
- We chose the color story (red = CEO, blue = worker, purple = company, gray = sector) after noticing that blue dots could be misread as worker pay. We also asked for a hover outline on the dots.
- We critically reviewed each commit locally before pushing (human reader and second agent critique) and checked the cleaned data against our Milestone 7 numbers (Amazon 4453 BCE, 34 companies below the poverty line, median 10.9 work hours)
    - With the cleaned data, we caught one wrong number in Claude's summary (it should be 190:1, not 186:1)

The CEO Pay Gap

In 2021, **Amazon's CEO** earned **6,474 times** what the company's **median worker** earned. Watch that gap grow while you read.

Since you opened this page (60 seconds ago):

Amazon's CEO earned

$408.13

Amazon's median worker earned

$0.06

Based on 2021 pay: **$212,701,169.00** for the CEO and **$32,855.00** for the median worker.

Every company tells a version of this story. Scroll to see where yours fits.

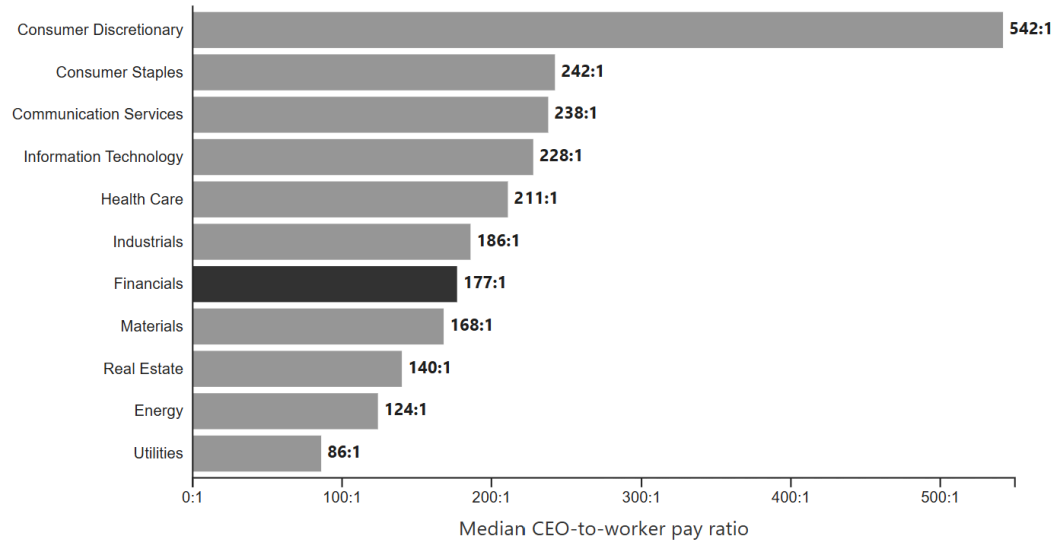
When would you have had to start working?

A company's pay ratio is also the number of years its **median worker** would need to work to earn what the **CEO** made in one year.

Which sectors have the biggest gaps?

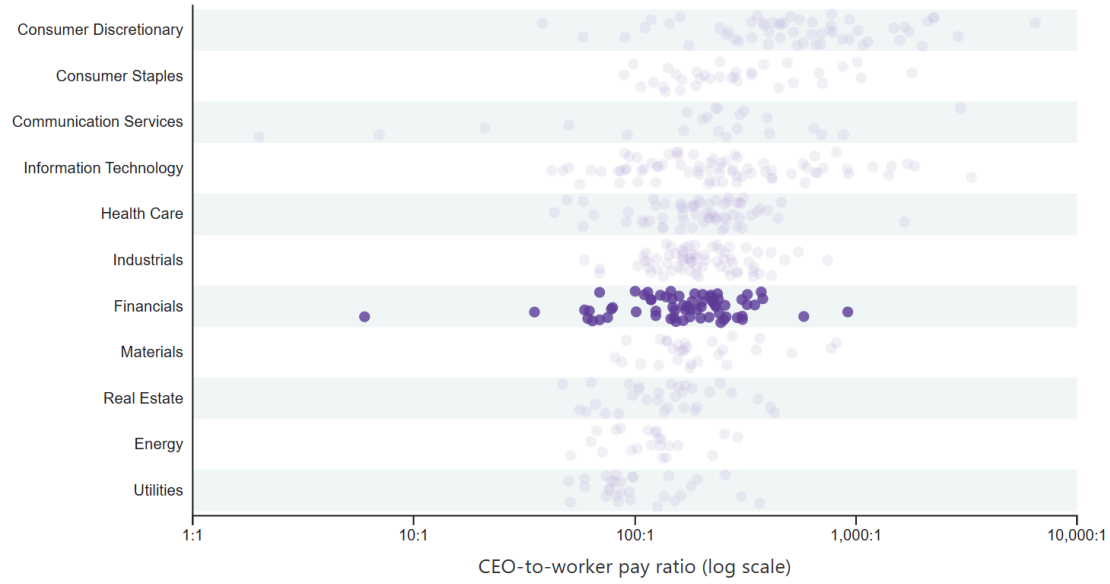
Every sector has a large gap, but the companies inside each sector differ even more.

Hover a sector to highlight its companies. Click a sector to show only it.



> Showing **492** companies in **all sectors** · median pay ratio **190:1**

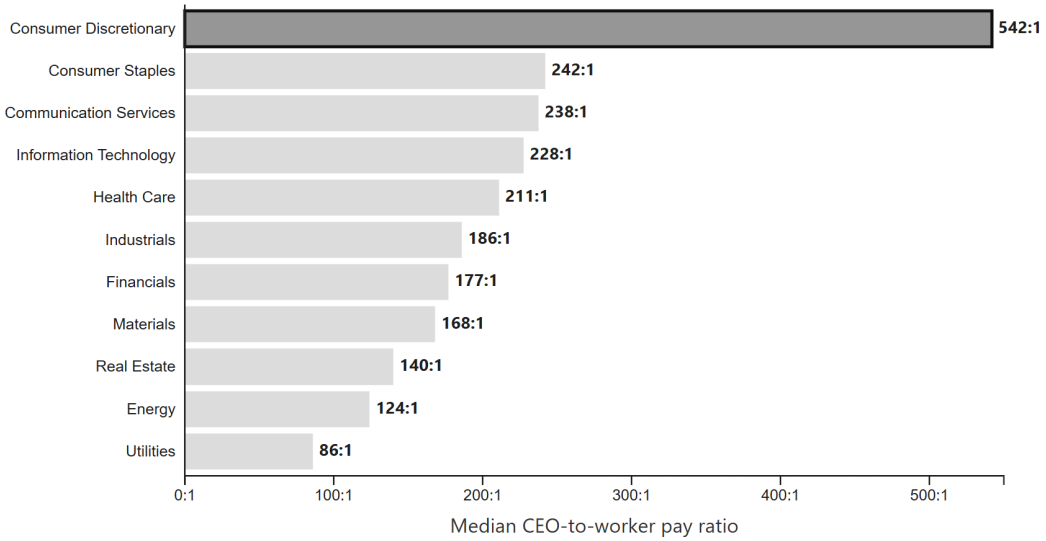
Each dot is one company. Hover a dot for its pay details.



Which sectors have the biggest gaps?

Every sector has a large gap, but the companies inside each sector differ even more.

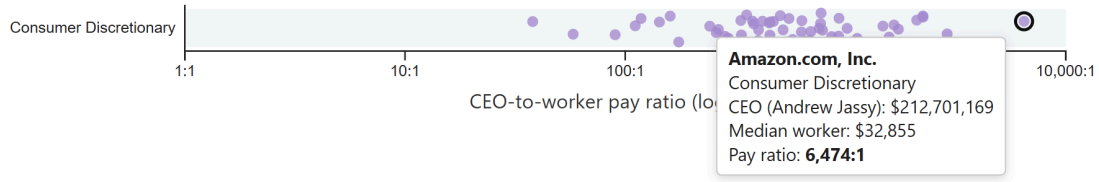
Hover a sector to highlight its companies. Click a sector to show only it.



Showing **55** companies in **Consumer Discretionary** · median pay ratio **542:1** ·

Show all sectors

Each dot is one company. Hover a dot for its pay details.



Is it high pay at the top, or low pay at the bottom?

The companies with the most extreme ratios are not always the ones with the **highest-paid CEOs**. Sometimes the gap comes from **very low worker pay**.

Each dot is one company. Dots above a dashed line have a higher pay ratio than that line. Hover a dot for its pay details. Click a dot to see how long its CEO takes to earn a worker's year.

Take-Two Interactive Software, Inc.

130.0 workdays

is how long it took CEO Strauss Zelnick to earn what the company's median worker earns in a full year (**$70,096**).

Pay ratio 2:1: the median worker would need about **2 years** to earn the CEO's pay for one year.



Below the poverty line

At **34** S&P 500 companies, the **median worker** earns less than the poverty line for a family of four.

COMING SOON

Below-the-poverty-line callout

Shows: The 34 companies whose median worker earned under $26,500 (2021 poverty line, family of four) as a grid of cards, with each CEO's pay next to the worker's. 19 of the 34 are in Consumer Discretionary.

Interaction: Filter the cards by sector.

Get your company's pay receipt

Pick a company and see when its CEO "earned" a year of your pay.

COMING SOON · NOVEL VIEW

Your company's pay receipt

Shows: A store-style receipt for the chosen company: CEO pay vs. median worker pay, the time on Jan 1 when the CEO had already "earned" the worker's year, and the year the worker would have had to start working to match the CEO.

Interaction: Search for a company to print its receipt, then download or share it as an image for campaign material.

Milestone 9 | Prototype II

Live link: https://aisharashidahmed.github.io/CS1710-Final_Project

Code: https://github.com/aisharashidahmed/CS1710-Final_Project/tree/main

TF Feedback

- ☐ Would like a title on the Timeline visualization
- ☐ See more visuals, NOT just numbers (more visual representation)
- ☐ Make the cards visual (right now it is all just numbers)
- ☐ Make visualizations more intuitive
- ☐ Define Pay Ratio early (at top) IN BIG LETTERS
- ☐ Revise the language to make it NOT AI-like (fix wording)

In-class Testing Feedback (Nov 4)

What the tester said	What we changed
Clicking dots was **not obvious** or intuitive in the scatterplot.	Added text and visual cursor cues to encourage clicking dots for details.
The **log scale** made it **difficult to interpret** exact pay ratio differences visually.	Included reference lines to clarify values on logarithmic axes. Also added a regular linear scale user can switch between.
Consumer Discretionary sector had too wide a spread of points, causing clutter.	Adjusted padding and spacing in sector views to reduce visual clutter.
Overlapping points made individual company selection frustrating.	Implemented subtle jitter and stroke highlights to improve point distinction. Lowered opacity and added a white outline.
The single-column layout (text then visualization below, then more text, etc) for cards felt long and repetitive.	Reorganized key metrics into a side-by-side card layout for better readability and allows for easier connection/referencing between story and data.
It wasn't clear which company was selected as the default example.	Pre-highlighted Amazon by default to provide an immediate example.
Users wanted to focus closer on low gap companies in the timeline instead of the full visual.	Started from zoom-in and slowly scaled out, and allowed scroll for continuation of story.

What changed since Milestone 8?
Key changes since Milestone 8 include converting the timeline artifact to a scroll story, starting from low pay-gap companies like Berkshire in recent years on the timeline, where scrolling zooms outward to then show the typical pay-gap, then the Amazon outlier. Testers commented on line breaks not matching with visualizations, so we made the text and visualizations wider and larger for better visibility. They also commented on not understanding the purpose/context of the website, so an intro paragraph was included. Aesthetics were also included to keep away from bland white. Additionally, more sections were implemented, but need to be revised.

Data Correction: A small bug was found when I (Ash) identified that there does not exist a year 0, so companies with a large enough pay gap to send their dots in the timeline within BCE years were 1-year off, and only needed an extra year to be added to their start date. For example, previously Amazon's start year was 4453 BCE, but it is now corrected to 4454 BCE.